Phase 1

In [2]:
```python
import pandas as pd

print("=== PHASE 1: DATE EXTRACTION & SORTING ===\n")

file_path = 'global_fuel_prices_2020_2026.csv'
df = pd.read_csv(file_path)

# Convert date and SORT DATA (Crucial for Time-Series Lags!)
df['date'] = pd.to_datetime(df['date'])
if 'country' in df.columns:
    df = df.sort_values(by=['country', 'date']).reset_index(drop=True)

# Separate date dataframe
df_dates = pd.DataFrame()
df_dates['date'] = df['date']

# Temporal features
df_dates['month'] = df_dates['date'].dt.month
df_dates['year'] = df_dates['date'].dt.year
df_dates['quarter'] = 'Q' + df_dates['date'].dt.quarter.astype(str)
df_dates['week_of_year'] = df_dates['date'].dt.isocalendar().week.astype(int)

# Remove date from main df
df = df.drop(columns=['date'])

print(f"Dataset Shape: {df.shape}")
print(f"Date Range: {df_dates['date'].min()} to {df_dates['date'].max()}")
display(df_dates.head())
```

```
=== PHASE 1: DATE EXTRACTION & SORTING ===

Dataset Shape: (27468, 9)
Date Range: 2020-01-06 00:00:00 to 2026-04-06 00:00:00
```

	date	month	year	quarter	week_of_year
0	2020-01-06	1	2020	Q1	2
1	2020-01-13	1	2020	Q1	3
2	2020-01-20	1	2020	Q1	4
3	2020-01-27	1	2020	Q1	5
4	2020-02-03	2	2020	Q1	6

In [3]:
```python
print("=== PHASE A: EXPLICIT CHRONOLOGICAL TRAIN/TEST SPLIT ===\n")

# Single source of truth for the time-series split. df_dates is row-aligned to df
# (both were sorted/reset together in Phase 1), so this boolean mask applies to
# df, df_dates, df_clean, df_model interchangeably.
# Reproducibility fix: train_mask was previously USED in Phase 4 & 5 but never
# DEFINED -> "Restart & Run All" crashed with NameError. Define it once here.
SPLIT_DATE = '2025-04-01'
train_mask = (df_dates['date'] < SPLIT_DATE).to_numpy()
test_mask = ~train_mask

print(f"Split date           : {SPLIT_DATE}")
print(f"Train rows           : {train_mask.sum()}  ({train_mask.mean():.1%})")
print(f"Test  rows           : {test_mask.sum()}  ({test_mask.mean():.1%})")
print(f"Train date range     : {df_dates['date'][train_mask].min().date()} -> {df_dates['date'][train_mask].max().date()}")
print(f"Test  date range     : {df_dates['date'][test_mask].min().date()} -> {df_dates['date'][test_mask].max().date()}")
```

```
=== PHASE A: EXPLICIT CHRONOLOGICAL TRAIN/TEST SPLIT ===

Split date           : 2025-04-01
Train rows           : 23016  (83.8%)
Test  rows           : 4452  (16.2%)
Train date range     : 2020-01-06 -> 2025-03-31
Test  date range     : 2025-04-07 -> 2026-04-06
```

Phase 2

In [4]:
```python
print("=== PHASE 2: SEASON ASSIGNMENT ===\n")

def assign_season_details(region, month):
    if region in ['Europe', 'North America', 'Asia']:
        group = 'Model A (4-Seasons)'
        if month in [12,1,2]: s='Winter'
        elif month in [3,4,5]: s='Spring'
        elif month in [6,7,8]: s='Summer'
        else: s='Autumn'

    elif region in ['South America','Oceania']:
        group='Model A (4-Seasons)'
        if month in [12,1,2]: s='Summer'
        elif month in [3,4,5]: s='Autumn'
        elif month in [6,7,8]: s='Winter'
        else: s='Spring'

    elif region=='Middle East':
        group='Model B (2-Seasons)'
        s='Hot' if 4 <= month <= 10 else 'Mild'

    elif region=='Africa':
        group='Model B (2-Seasons)'
        s='Wet' if 5 <= month <=10 else 'Dry'

    else:
        group='Unknown'
        s='Unknown'

    return s, group

temp_df = pd.DataFrame({'region': df['region'], 'month': df_dates['month']})

df[['season','region_group']] = temp_df.apply(
    lambda row: pd.Series(assign_season_details(row['region'], row['month'])), axis=1
)

print(df['season'].value_counts())
```

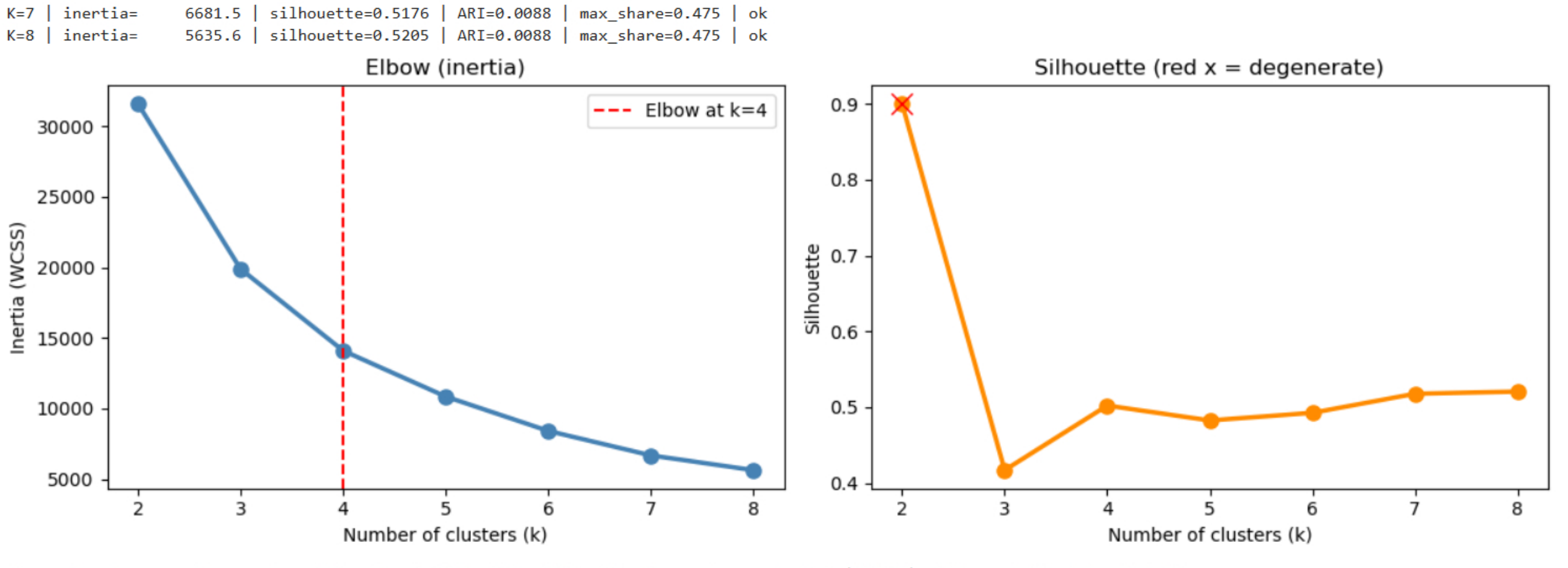
```
=== PHASE 2: SEASON ASSIGNMENT ===

season
Winter    4366
Spring    4357
Summer    4162
Autumn    4119
Dry       3887
Wet       3634
Hot       1656
Mild      1287
Name: count, dtype: int64
season
Winter    4366
Spring    4357
Summer    4162
Autumn    4119
Dry       3887
Wet       3634
Hot       1656
Mild      1287
Name: count, dtype: int64
```

In [5]:
```python
from sklearn.cluster import KMeans
import numpy as np

print("=== PHASE 2.5: EARLY WARNING SYSTEM (EWS) ALERTS ===\n")

# 1. Calculate Cross-Commodity Ratio
df['cross_ratio'] = df['diesel_usd_liter'] / df['petrol_usd_liter']

# 2. Calculate Brent Momentum (4-Week % Change)
# Update: Using your exact column name 'brent_crude_usd'
df['brent_momentum'] = df.groupby('country')['brent_crude_usd'].pct_change(periods=4)

# 3. THE LAG TRICK (Prevention of Data Leakage)
# Hum aaj ka ratio use nahi kar sakte, sirf pichle hafte (Lag 1) ka use karenge!
df['cross_ratio_lag1'] = df.groupby('country')['cross_ratio'].shift(1)
df['brent_momentum_lag1'] = df.groupby('country')['brent_momentum'].shift(1)

# 4. Drop current ratio and momentum so model doesn't cheat (Data Leakage proofing)
df = df.drop(columns=['cross_ratio', 'brent_momentum'])

# 5. Build clustering features and SCALE them.
# Problem #3 fix: K-Means is distance-based, so it MUST run on standardized features.
# On raw features (ratio ~1, momentum ~0..big) one dimension dominated and 96.6% of
# weeks collapsed into a single cluster. We StandardScale and fit on TRAIN rows only.
from sklearn.preprocessing import StandardScaler

cluster_feats = ['cross_ratio_lag1', 'brent_momentum_lag1']

# median imputation fit on TRAIN rows only (leak-free), then applied to all rows
cluster_medians = df.loc[train_mask, cluster_feats].median()
cluster_data = df[cluster_feats].fillna(cluster_medians)

ews_scaler = StandardScaler().fit(cluster_data[train_mask])
cluster_scaled = ews_scaler.transform(cluster_data)
cluster_scaled_train = cluster_scaled[train_mask]

print("Cluster features standardized (StandardScaler fit on train rows only).")
print(f"Train means ~0: {cluster_scaled_train.mean(axis=0).round(3)} | "
      f"Train stds ~1: {cluster_scaled_train.std(axis=0).round(3)}")
print("Final cluster fit + EWS_Alert_Level assignment happens after K-selection below.")
```

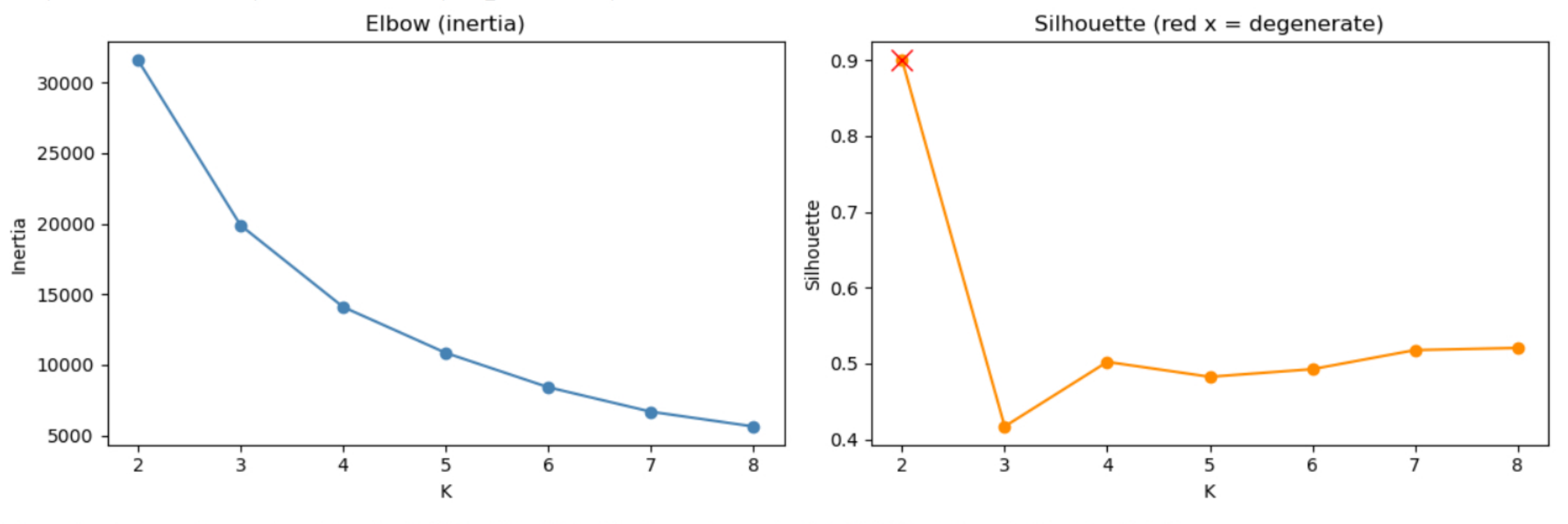
```
=== PHASE 2.5: EARLY WARNING SYSTEM (EWS) ALERTS ===

Cluster features standardized (StandardScaler fit on train rows only).
Train means ~0: [ 0. -0.] | Train stds ~1: [1. 1.]
Final cluster fit + EWS_Alert_Level assignment happens after K-selection below.
Cluster features standardized (StandardScaler fit on train rows only).
Train means ~0: [ 0. -0.] | Train stds ~1: [1. 1.]
Final cluster fit + EWS_Alert_Level assignment happens after K-selection below.
```

In [6]:
```python
from sklearn.metrics import silhouette_score, adjusted_rand_score
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from sklearn.cluster import KMeans

print("=== PHASE C: K-SELECTION SWEEP (silhouette + elbow) ===\n")

# Justify K instead of hardcoding 4. Sweep K = 2..8 on the TRAIN scaled features.
# Silhouette is scored on a fixed random subsample for speed/stability (n large).
# ARI is compared against an objective shock label derived from the same petrol-change
# rule used later in Phase E, so the sweep can evaluate cluster structure vs ground truth.
K_RANGE = range(2, 9)
rng = np.random.RandomState(42)
n_sample = min(5000, cluster_scaled_train.shape[0])
sample_idx = rng.choice(cluster_scaled_train.shape[0], size=n_sample, replace=False)
sample = cluster_scaled_train[sample_idx]

DEGEN_SHARE = 0.90   # a solution is "degenerate" if one cluster holds >90% of rows

# Ground-truth shock labels aligned to df row order.
# This mirrors the Phase E shock definition but is local to Phase C so the sweep does not
# depend on later cells.
ews_truth = pd.DataFrame({
    'country': df['country'].to_numpy(),
    'date': df_dates['date'].to_numpy(),
    'petrol': df['petrol_usd_liter'].to_numpy(),
}).sort_values(['country', 'date']).reset_index(drop=True)

g_truth = ews_truth.groupby('country')
ews_truth['chg'] = g_truth['petrol'].diff()
ews_truth['sigma'] = g_truth['chg'].transform(lambda s: s.shift(1).rolling(26, min_periods=8).std())
ews_truth['shock'] = (ews_truth['chg'].abs() > 2.0 * ews_truth['sigma']) & ews_truth['sigma'].notna()
shock_truth = ews_truth['shock'].to_numpy()

sweep = []
for k in K_RANGE:
    km = KMeans(n_clusters=k, random_state=42, n_init=10).fit(cluster_scaled_train)
    sil = silhouette_score(sample, km.predict(sample))
    sizes = np.bincount(km.labels_, minlength=k)
    max_share = sizes.max() / sizes.sum()
    degenerate = max_share > DEGEN_SHARE
    ari = adjusted_rand_score(shock_truth[train_mask], km.predict(cluster_scaled)[train_mask])
    sweep.append({'K': k, 'inertia': km.inertia_, 'silhouette': sil,
                  'ari': ari, 'max_cluster_share': max_share, 'degenerate': degenerate})
    flag = 'DEGENERATE (one cluster >90%)' if degenerate else 'ok'
    print(f"K={k} | inertia={km.inertia_:11.1f} | silhouette={sil:.4f} | ARI={ari:.4f} | "
          f"max_share={max_share:.3f} | {flag}")

sweep_df = pd.DataFrame(sweep)
sil_by_k = dict(zip(sweep_df['K'], sweep_df['silhouette']))

# IMPORTANT: the global silhouette max is at K=2, but that solution is DEGENERATE -
# it only isolates ~130 petrol-subsidy rows (Venezuela/Iran/Libya, petrol ~$0.01/L,
# diesel/petrol ratio 3-7x) from everything else. So the raw silhouette winner is an
# outlier artifact, not a market regime. We therefore choose K among NON-degenerate
# solutions and, since silhouette is flat there (~0.48-0.52), prefer K=4 for an
# interpretable 4-level alert ladder.
valid = sweep_df[~sweep_df['degenerate']]
best_valid_k = int(valid.loc[valid['silhouette'].idxmax(), 'K'])

best_k = int(sweep_df.loc[sweep_df['silhouette'].idxmax(), 'K'])
fig, ax = plt.subplots(1, 2, figsize=(12, 4))

# -- ELBOW PLOT MODIFICATIONS --
ax[0].plot(sweep_df['K'], sweep_df['inertia'], 'o-', color='steelblue', linewidth=2.5, markersize=8)
ax[0].set(title='Elbow (inertia)', xlabel='Number of clusters (k)', ylabel='Inertia (WCSS)')

# Vertical straight dashed line add karne ka code
ax[0].axvline(x=4, color='red', linestyle='--', label='Elbow at k=4')
ax[0].legend()
# ------------------------------

ax[1].plot(sweep_df['K'], sweep_df['silhouette'], 'o-', color='darkorange', linewidth=2.5, markersize=8)
for _, r in sweep_df.iterrows():
    if r['degenerate']:
        ax[1].plot(r['K'], r['silhouette'], 'x', color='red', ms=12)
ax[1].set(title='Silhouette (red x = degenerate)', xlabel='Number of clusters (k)', ylabel='Silhouette')
plt.tight_layout(); plt.show()

if (4 in valid['K'].values) and (sil_by_k[best_valid_k] - sil_by_k[4] <= 0.05):
    CHOSEN_K = 4
    print(f"\nChosen K = 4 -> non-degenerate, silhouette {sil_by_k[4]:.4f} within 0.05 of "
          f"best non-degenerate K={best_valid_k} ({sil_by_k[best_valid_k]:.4f}); interpretable 4-level ladder.")
else:
    CHOSEN_K = best_valid_k
    print(f"\nChosen K = {best_valid_k} (best non-degenerate silhouette {sil_by_k[best_valid_k]:.4f}; "
          f"K=4 silhouette {sil_by_k.get(4):.4f}).")
```

```
=== PHASE C: K-SELECTION SWEEP (silhouette + elbow) ===

K=2 | inertia=    31577.8 | silhouette=0.9002 | ARI=-0.0013 | max_share=0.995 | DEGENERATE (one cluster >90%)
K=3 | inertia=    19886.1 | silhouette=0.4169 | ARI=-0.0020 | max_share=0.574 | ok
K=4 | inertia=    14097.8 | silhouette=0.5021 | ARI=0.0121 | max_share=0.672 | ok
K=5 | inertia=    10849.0 | silhouette=0.4824 | ARI=0.0086 | max_share=0.487 | ok
K=6 | inertia=     8413.1 | silhouette=0.4925 | ARI=0.0084 | max_share=0.485 | ok
K=7 | inertia=     6681.5 | silhouette=0.5176 | ARI=0.0088 | max_share=0.475 | ok
K=8 | inertia=     5635.6 | silhouette=0.5205 | ARI=0.0088 | max_share=0.475 | ok
```

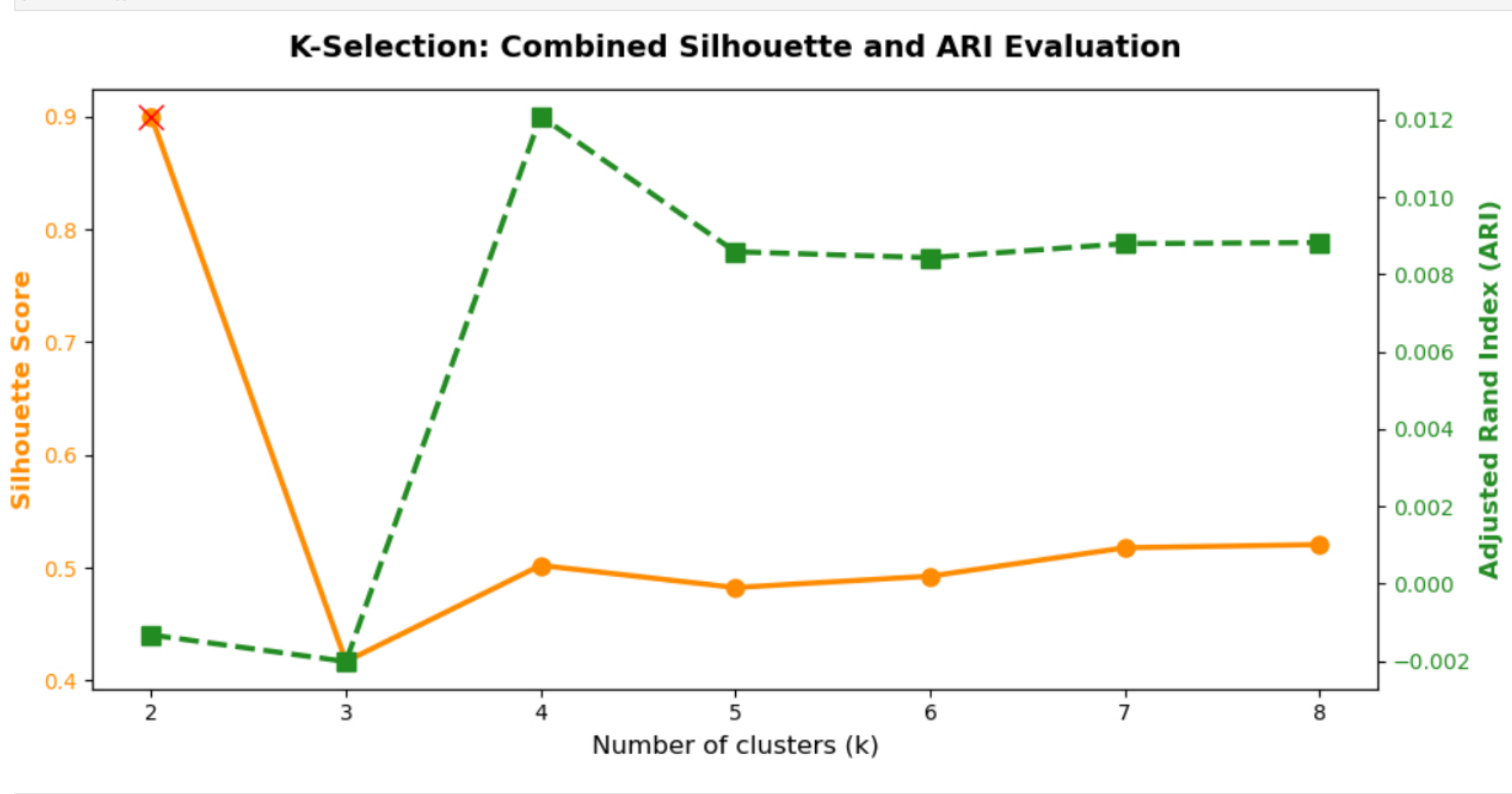


```
Chosen K = 4 -> non-degenerate, silhouette 0.5021 within 0.05 of best non-degenerate K=8 (0.5205); interpretable 4-level ladder.
K=8 | inertia=     5635.6 | silhouette=0.5205 | max_share=0.475 | ok
```



```
Chosen K = 4 -> non-degenerate, silhouette 0.5021 within 0.05 of best non-degenerate K=8 (0.5205); interpretable 4-level ladder.
```

In [7]:
```python
import matplotlib.pyplot as plt

fig, ax1 = plt.subplots(figsize=(10, 5))

# First Plot: Silhouette Score (Left Y-Axis)
color1 = 'darkorange'
ax1.set_xlabel('Number of clusters (k)', fontsize=12)
ax1.set_ylabel('Silhouette Score', color=color1, fontsize=12, fontweight='bold')
ax1.plot(sweep_df['K'], sweep_df['silhouette'], 'o-', color=color1, linewidth=2.5, markersize=8, label='Silhouette')
ax1.tick_params(axis='y', labelcolor=color1)

# Second Plot: ARI Score (Right Y-Axis)
ax2 = ax1.twinx()
color2 = 'forestgreen'
ax2.set_ylabel('Adjusted Rand Index (ARI)', color=color2, fontsize=12, fontweight='bold')
ax2.plot(sweep_df['K'], sweep_df['ari'], 's--', color=color2, linewidth=2.5, markersize=8, label='ARI')
ax2.tick_params(axis='y', labelcolor=color2)

# Highlight degenerate clusters (Red X marks)
for _, r in sweep_df.iterrows():
    if r['degenerate']:
        ax1.plot(r['K'], r['silhouette'], 'x', color='red', ms=12)

# Title and layout
plt.title('K-Selection: Combined Silhouette and ARI Evaluation', fontsize=14, fontweight='bold', pad=15)
fig.tight_layout()
plt.savefig('silhouette_ari_combined.png', dpi=300, bbox_inches='tight')
plt.show()
```



In [6]:
```python
print("=== PHASE C: FINAL CLUSTERING + DERIVED REGIME LABELS ===\n")

# Fit final KMeans on TRAIN scaled features, then assign labels to ALL rows.
ews_kmeans = KMeans(n_clusters=CHOSEN_K, random_state=42, n_init=10).fit(cluster_scaled_train)
raw_cluster = ews_kmeans.predict(cluster_scaled)

# Centroids back in ORIGINAL units (auditable).
centroids_orig = ews_scaler.inverse_transform(ews_kmeans.cluster_centers_)
cent_df = pd.DataFrame(centroids_orig, columns=cluster_feats)
cent_df['n_rows'] = pd.Series(raw_cluster).value_counts().sort_index()

# DERIVE the regime ordering from centroids instead of assuming 0=Normal..3=Crisis.
# Alert severity = price-shock pressure, so we rank PRIMARILY by centroid Brent
# momentum (higher = more inflationary stress). Ratio extremity (distance of the
# diesel/petrol ratio from normal) is only a tiebreak; on its own it mostly flags
# static petrol-subsidy states, which are not "crises".
ratio_norm = cluster_data.loc[train_mask, 'cross_ratio_lag1'].median()
cent_df['ratio_extremity'] = (cent_df['cross_ratio_lag1'] - ratio_norm).abs()
order = (cent_df.sort_values(['brent_momentum_lag1', 'ratio_extremity'])
             .index.tolist())  # calm -> severe

if CHOSEN_K == 4:
    regime_names = ['Normal', 'Local Peak', 'Global Inflation', 'Severe Crisis']
else:
    regime_names = [f'Level {i}' for i in range(CHOSEN_K)]
    regime_names[0] += ' (calmest)'; regime_names[-1] += ' (most severe)'

# map raw cluster id -> ordered alert level (0 = calm) and a human regime name
cluster_to_level = {raw_id: lvl for lvl, raw_id in enumerate(order)}
level_to_name = {lvl: regime_names[lvl] for lvl in range(CHOSEN_K)}

df['EWS_Alert_Level'] = pd.Series(raw_cluster, index=df.index).map(cluster_to_level)
df['EWS_Regime'] = df['EWS_Alert_Level'].map(level_to_name)

cent_df['alert_level'] = cent_df.index.map(cluster_to_level)
cent_df['regime'] = cent_df['alert_level'].map(level_to_name)
cent_df = cent_df.sort_values('alert_level')

print("Centroid table (original units) -> mapping is auditable, NOT assumed:")
display(cent_df[['regime', 'alert_level', 'cross_ratio_lag1', 'brent_momentum_lag1',
                 'ratio_extremity', 'n_rows']].round(4))

print("\nNew (non-degenerate) cluster sizes by alert level:")
print(df['EWS_Alert_Level'].value_counts().sort_index())
print("\nRegime distribution:")
print(df['EWS_Regime'].value_counts())
```

```
=== PHASE C: FINAL CLUSTERING + DERIVED REGIME LABELS ===

Centroid table (original units) -> mapping is auditable, NOT assumed:
```

	regime	alert_level	cross_ratio_lag1	brent_momentum_lag1	ratio_extremity	n_rows
3	Normal	0	0.9349	-0.1490	0.0065	668
2	Local Peak	1	0.9238	-0.0060	0.0045	19150
1	Global Inflation	2	3.2306	0.0103	2.3023	127
0	Severe Crisis	3	0.9282	0.0616	0.0001	7523

```
New (non-degenerate) cluster sizes by alert level:
EWS_Alert_Level
0      668
1    19150
2      127
3     7523
Name: count, dtype: int64

Regime distribution:
EWS_Regime
Local Peak          19150
Severe Crisis        7523
Normal                668
Global Inflation      127
Name: count, dtype: int64
```

Phase 3

In [7]:
```python
import numpy as np
import pandas as pd

print("=== PHASE 3: DATA CLEANING ===\n")

df_cat = df.select_dtypes(exclude=[np.number]).copy()
df_num = df.select_dtypes(include=[np.number]).copy()

# Leakage fix: impute using statistics computed on TRAIN ROWS ONLY, then apply to
# train+test. Previously mode()/median() were fit on the full dataset (incl. test).
for col in df_cat.columns:
    fill_val = df_cat[col][train_mask].mode()[0]
    df_cat[col] = df_cat[col].fillna(fill_val)

for col in df_num.columns:
    fill_val = df_num[col][train_mask].median()
    df_num[col] = df_num[col].fillna(fill_val)

# Recombine cleaned data
df_clean = pd.concat([df_cat, df_num], axis=1)
df_clean = df_clean[df.columns]

print("Data Cleaned Successfully ✅")
print(f"Cleaned Data Shape: {df_clean.shape}")
```

```
=== PHASE 3: DATA CLEANING ===

Data Cleaned Successfully ✅
Cleaned Data Shape: (27468, 15)
```

Phase 4

In [8]:
```python
from sklearn.preprocessing import LabelEncoder, StandardScaler

print("=== PHASE 4: FEATURE ENGINEERING + ENCODING ===\n")

df_model = df_clean.copy()

# EWS_Regime is a human-readable label for interpretability/plots only; the numeric
# EWS_Alert_Level is the model feature. Drop the string column so it never enters X.
df_model = df_model.drop(columns=['EWS_Regime'], errors='ignore')

# Add temporal features back
df_model['month'] = df_dates['month']
df_model['year'] = df_dates['year']
df_model['week_of_year'] = df_dates['week_of_year']

# Label Encoding
# Leakage fix: fit the encoder mapping on TRAIN ROWS ONLY, then apply to all rows.
# Unseen test-only categories map to -1 (none exist in this dataset, but this keeps
# a fresh run robust). Previously fit_transform ran on the full column (incl. test).
categorical_cols = ['season', 'region', 'region_group', 'income_level', 'subsidy_level']
for col in categorical_cols:
    if col in df_model.columns:
        full_col = df_model[col].astype(str)
        le = LabelEncoder()
        le.fit(full_col[train_mask])
        mapping = {cat: idx for idx, cat in enumerate(le.classes_)}
        df_model[col] = full_col.map(mapping).fillna(-1).astype(int)

# Country One Hot Encoding
if 'country' in df_model.columns:
    df_model = pd.get_dummies(df_model, columns=['country'], prefix='country')

country_ohe_cols = [c for c in df_model.columns if c.startswith('country_')]
df_model[country_ohe_cols] = df_model[country_ohe_cols].astype(int)

# Train/Test Split
X = df_model.drop(columns=['diesel_usd_liter', 'petrol_usd_liter', 'lpg_usd_liter'], errors='ignore')

# Hum Fuel Targets baad mein define karenge Phase 6 mein, yahan sirf general split karte hain
X_train_raw = X[train_mask]
X_test_raw = X[~train_mask]

# Clean remaining objects
obj_cols = X_train_raw.select_dtypes(include=['object']).columns
if len(obj_cols) > 0:
    print("Still object columns found:", list(obj_cols))
else:
    print("No object columns found ✅")

print("\n=== PHASE 4 COMPLETE ===")
print(f"Training Shape: {X_train_raw.shape}")
print(f"Testing Shape: {X_test_raw.shape}")
```

```
=== PHASE 4: FEATURE ENGINEERING + ENCODING ===

No object columns found ✅

=== PHASE 4 COMPLETE ===
Training Shape: (23016, 97)
Testing Shape: (4452, 97)
```

Phase 5

In [9]:
```python
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.neighbors import KNeighborsRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import TimeSeriesSplit, cross_val_score
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from xgboost import XGBRegressor

print("=== PHASE 5 & 6: FINAL MODEL TRAINING ===\n")

fuel_targets = ['diesel_usd_liter', 'petrol_usd_liter', 'lpg_usd_liter']
results = []

def evaluate_model(model, Xtr, Xte, ytr, yte, name, fuel):
    model.fit(Xtr, ytr)
    pred = model.predict(Xte)
    r2 = r2_score(yte, pred)
    mae = mean_absolute_error(yte, pred)
    rmse = np.sqrt(mean_squared_error(yte, pred))

    tscv = TimeSeriesSplit(n_splits=5)
    cv = cross_val_score(model, Xtr, ytr, cv=tscv, scoring='r2').mean()

    results.append([fuel, name, cv, r2, mae, rmse])
    print(f"{name} -> CV R2: {cv:.4f} | Test R2: {r2:.4f} | RMSE: {rmse:.4f}")

for fuel in fuel_targets:
    if fuel not in df_model.columns:
        continue

    print(f"\n========== {fuel.upper()} ==========\n")
    y = df_model[fuel]
    y_train = y[train_mask]
    y_test = y[~train_mask]

    # Target aur baqi fuels drop kar dein taake Data Leakage na ho
    X_fuel = df_model.drop(columns=['diesel_usd_liter', 'petrol_usd_liter', 'lpg_usd_liter'], errors='ignore')
    X_train_fuel = X_fuel[train_mask]
    X_test_fuel = X_fuel[~train_mask]

    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_train_fuel)
    X_test_scaled = scaler.transform(X_test_fuel)

    evaluate_model(LinearRegression(), X_train_scaled, X_test_scaled, y_train, y_test, "Linear Regression", fuel)
    evaluate_model(KNeighborsRegressor(n_neighbors=5), X_train_scaled, X_test_scaled, y_train, y_test, "KNN", fuel)
    evaluate_model(RandomForestRegressor(n_estimators=150, max_depth=12, random_state=42), X_train_fuel, X_test_fuel, y_train, y_test, "Random Forest", fuel)
    evaluate_model(XGBRegressor(n_estimators=150, max_depth=6, learning_rate=0.1, random_state=42), X_train_fuel, X_test_fuel, y_train, y_test, "XGBoost", fuel)

# Final Results
results_df = pd.DataFrame(results, columns=['Fuel', 'Model', 'CV_R2', 'Test_R2', 'MAE', 'RMSE'])
print("\n=== FINAL COMPARISON TABLE ===")
display(results_df.sort_values(by=['Fuel', 'Test_R2'], ascending=[True, False]))
```

```
=== PHASE 5 & 6: FINAL MODEL TRAINING ===


========== DIESEL_USD_LITER ==========

Linear Regression -> CV R2: 0.2499 | Test R2: 0.9822 | RMSE: 0.2090
KNN -> CV R2: 0.7189 | Test R2: 0.9933 | RMSE: 0.1284
Random Forest -> CV R2: 0.6363 | Test R2: 0.9926 | RMSE: 0.1346
XGBoost -> CV R2: 0.7108 | Test R2: 0.9993 | RMSE: 0.0409

========== PETROL_USD_LITER ==========

Linear Regression -> CV R2: 0.2483 | Test R2: 0.9822 | RMSE: 0.2192
KNN -> CV R2: 0.7129 | Test R2: 0.9932 | RMSE: 0.1359
Random Forest -> CV R2: 0.6431 | Test R2: 0.9920 | RMSE: 0.1470
XGBoost -> CV R2: 0.7115 | Test R2: 0.9994 | RMSE: 0.0410

========== LPG_USD_LITER ==========

Linear Regression -> CV R2: 0.2484 | Test R2: 0.9820 | RMSE: 0.1654
KNN -> CV R2: 0.7128 | Test R2: 0.9929 | RMSE: 0.1037
Random Forest -> CV R2: 0.6483 | Test R2: 0.9918 | RMSE: 0.1116
XGBoost -> CV R2: 0.7127 | Test R2: 0.9992 | RMSE: 0.0356

=== FINAL COMPARISON TABLE ===
```

	Fuel	Model	CV_R2	Test_R2	MAE	RMSE
3	diesel_usd_liter	XGBoost	0.710795	0.999318	0.032551	0.040854
1	diesel_usd_liter	KNN	0.718933	0.993271	0.084364	0.128352
2	diesel_usd_liter	Random Forest	0.636333	0.992600	0.072722	0.134602
0	diesel_usd_liter	Linear Regression	0.249914	0.982163	0.170946	0.208970
11	lpg_usd_liter	XGBoost	0.712746	0.999167	0.028882	0.035573
9	lpg_usd_liter	KNN	0.712769	0.992926	0.069109	0.103679
10	lpg_usd_liter	Random Forest	0.648268	0.991808	0.061660	0.111571
8	lpg_usd_liter	Linear Regression	0.248432	0.981995	0.135531	0.165410
7	petrol_usd_liter	XGBoost	0.711486	0.999376	0.033646	0.041040
5	petrol_usd_liter	KNN	0.712878	0.993164	0.089160	0.135854
6	petrol_usd_liter	Random Forest	0.643086	0.991998	0.078044	0.146986
4	petrol_usd_liter	Linear Regression	0.248301	0.982205	0.179404	0.219195

Phase B — Persistence Baseline & Skill Score

The weekly fuel-price *levels* are ~99.95% autocorrelated week-to-week, so a model can score $R^2 \approx 0.999$ by essentially copying last week's price. To know whether the ML models add any real value we compare them against a **naive persistence baseline** (`price[t] = price[t-1]` per country) on the *same* test split, and report a **skill score**:

`skill = 1 - RMSE_model / RMSE_persistence` → positive means the model beats persistence.

```
In [10]: print("=== PHASE B: PERSISTENCE BASELINE & SKILL SCORE ===\n")

         # Naive persistence baseline: predict price[t] = price[t-1] WITHIN the same country.
         # This is the honest yardstick for highly autocorrelated weekly price levels.
         # Evaluated on EXACTLY the same test rows (test_mask) used by the ML models.
         persistence_rows = []
         persistence_rmse = {}   # fuel -> baseline RMSE, reused for the skill score

         for fuel in fuel_targets:
             # previous-week actual price within each country (uses only past data -> no leakage)
             prev = df.groupby('country')[fuel].shift(1)

             y_true = df.loc[test_mask, fuel].to_numpy()
             y_pred = prev[test_mask].to_numpy()

             # guard against any undefined lag (none expected inside the test block)
             valid = ~np.isnan(y_pred)
             y_true, y_pred = y_true[valid], y_pred[valid]

             r2 = r2_score(y_true, y_pred)
             mae = mean_absolute_error(y_true, y_pred)
             rmse = np.sqrt(mean_squared_error(y_true, y_pred))
             persistence_rmse[fuel] = rmse
             persistence_rows.append([fuel, 'Persistence (naive)', np.nan, r2, mae, rmse])
             print(f"{fuel:18s} persistence -> R2: {r2:.4f} | MAE: {mae:.4f} | RMSE: {rmse:.4f}")

         # Merge baseline rows into the model results and compute the skill score.
         baseline_df = pd.DataFrame(persistence_rows, columns=['Fuel', 'Model', 'CV_R2', 'Test_R2', 'MAE', 'RMSE'])
         results_full = pd.concat([results_df, baseline_df], ignore_index=True)

         results_full['Skill_Score'] = results_full.apply(
             lambda r: 1.0 - r['RMSE'] / persistence_rmse[r['Fuel']], axis=1
         )
         results_full['Beats_Persistence'] = np.where(
             results_full['Model'].eq('Persistence (naive)'), 'baseline',
             np.where(results_full['Skill_Score'] > 0, 'YES', 'NO')
         )
         results_full = results_full.sort_values(by=['Fuel', 'RMSE'], ascending=[True, True]).reset_index(drop=True)

         print("\n=== FINAL COMPARISON TABLE (models vs persistence + skill score) ===")
         display(results_full)

         print("\n=== NARRATIVE: does ML beat the naive baseline? ===")
         for fuel in fuel_targets:
             sub = results_full[(results_full['Fuel'] == fuel) & (results_full['Model'] != 'Persistence (naive)')]
             best = sub.loc[sub['RMSE'].idxmin()]
             pr = persistence_rmse[fuel]
             verdict = "BEATS" if best['Skill_Score'] > 0 else "does NOT beat"
             print(f"{fuel:18s}: persistence RMSE={pr:.4f} | best model={best['Model']} "
                   f"(RMSE={best['RMSE']:.4f}, skill={best['Skill_Score']:+.3f}) -> {verdict} persistence.")
```

```
=== PHASE B: PERSISTENCE BASELINE & SKILL SCORE ===

diesel_usd_liter   persistence -> R2: 0.9991 | MAE: 0.0339 | RMSE: 0.0464
petrol_usd_liter   persistence -> R2: 0.9991 | MAE: 0.0348 | RMSE: 0.0482
lpg_usd_liter      persistence -> R2: 0.9988 | MAE: 0.0317 | RMSE: 0.0423

=== FINAL COMPARISON TABLE (models vs persistence + skill score) ===
```

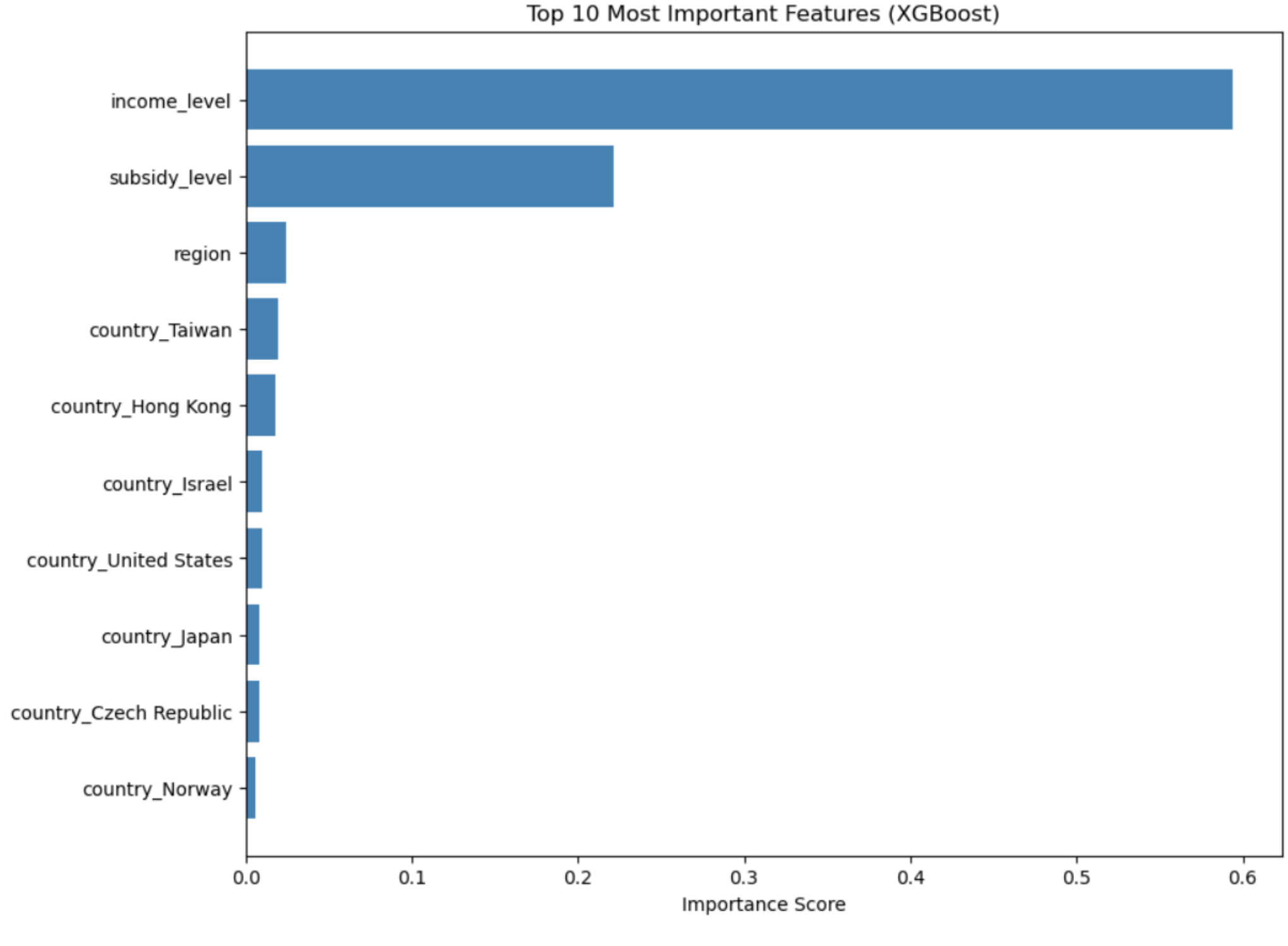
	Fuel	Model	CV_R2	Test_R2	MAE	RMSE	Skill_Score	Beats_Persistence
0	diesel_usd_liter	XGBoost	0.710795	0.999318	0.032551	0.040854	0.119017	YES
1	diesel_usd_liter	Persistence (naive)	NaN	0.999122	0.033861	0.046373	0.000000	baseline
2	diesel_usd_liter	KNN	0.718933	0.993271	0.084364	0.128352	-1.767811	NO
3	diesel_usd_liter	Random Forest	0.636333	0.992600	0.072722	0.134602	-1.902591	NO
4	diesel_usd_liter	Linear Regression	0.249914	0.982163	0.170946	0.208970	-3.506269	NO
5	lpg_usd_liter	XGBoost	0.712746	0.999167	0.028882	0.035573	0.159477	YES
6	lpg_usd_liter	Persistence (naive)	NaN	0.998821	0.031708	0.042322	0.000000	baseline
7	lpg_usd_liter	KNN	0.712769	0.992926	0.069109	0.103679	-1.449730	NO
8	lpg_usd_liter	Random Forest	0.648268	0.991808	0.061660	0.111571	-1.636221	NO
9	lpg_usd_liter	Linear Regression	0.248432	0.981995	0.135531	0.165410	-2.908321	NO
10	petrol_usd_liter	XGBoost	0.711486	0.999376	0.033646	0.041040	0.149164	YES
11	petrol_usd_liter	Persistence (naive)	NaN	0.999138	0.034785	0.048234	0.000000	baseline
12	petrol_usd_liter	KNN	0.712878	0.993164	0.089160	0.135854	-1.816543	NO
13	petrol_usd_liter	Random Forest	0.643086	0.991998	0.078044	0.146986	-2.047316	NO
14	petrol_usd_liter	Linear Regression	0.248301	0.982205	0.179404	0.219195	-3.544357	NO

```
=== NARRATIVE: does ML beat the naive baseline? ===
diesel_usd_liter  : persistence RMSE=0.0464 | best model=XGBoost (RMSE=0.0409, skill=+0.119) -> BEATS persistence.
petrol_usd_liter  : persistence RMSE=0.0482 | best model=XGBoost (RMSE=0.0410, skill=+0.149) -> BEATS persistence.
lpg_usd_liter     : persistence RMSE=0.0423 | best model=XGBoost (RMSE=0.0356, skill=+0.159) -> BEATS persistence.
```

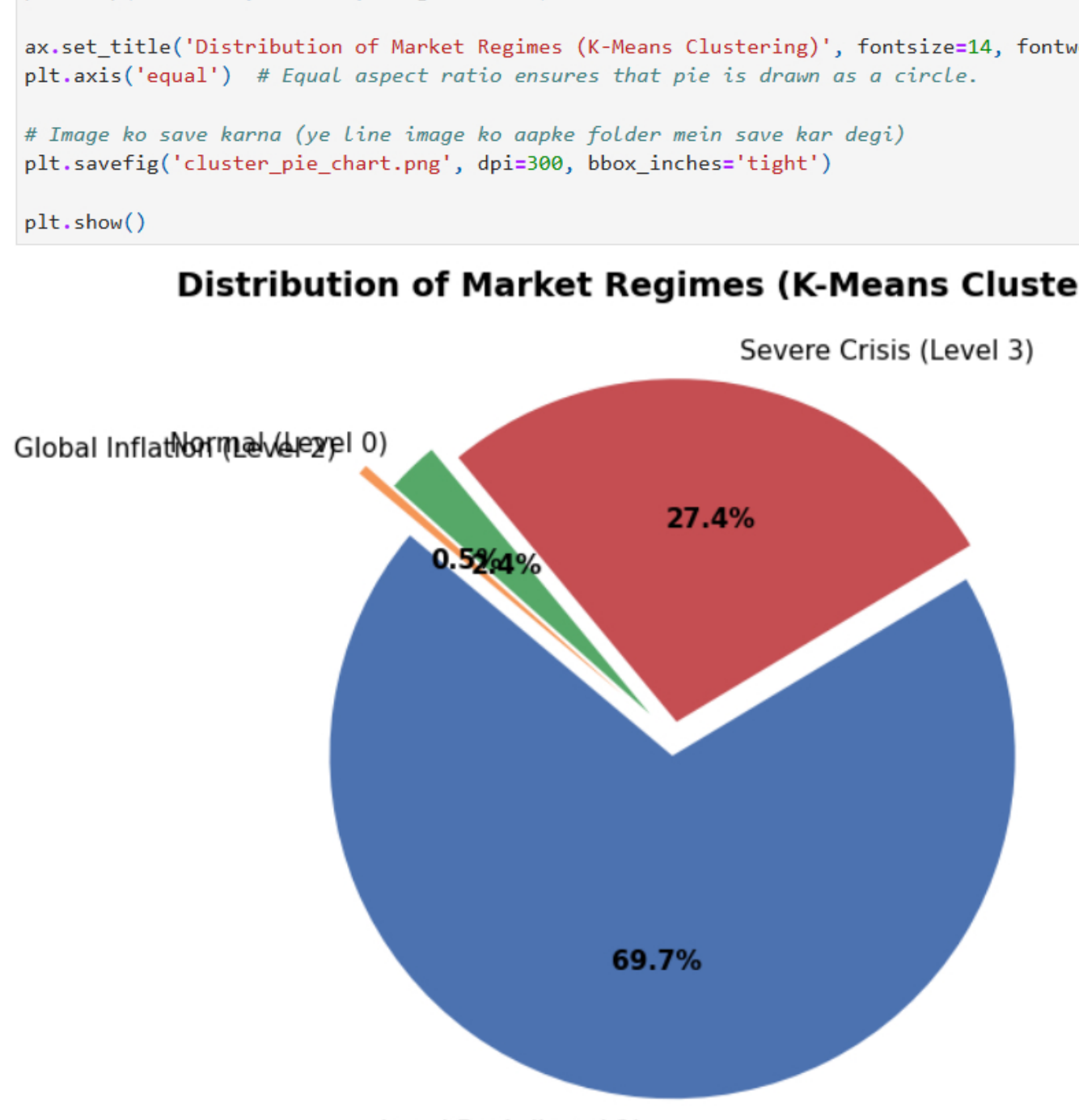
```
In [ ]:
```

```
In [11]: import matplotlib.pyplot as plt

         print("=== XGBOOST FEATURE IMPORTANCE ===")
         # Get the trained XGBoost model from your evaluation block
         # Assuming the last trained model in the loop was XGBoost for LPG
         xgb_model = XGBRegressor(n_estimators=150, max_depth=6, learning_rate=0.1, random_state=42)
         xgb_model.fit(X_train_raw, y_train)

         # Plotting
         importance = xgb_model.feature_importances_
         features = X_train_raw.columns

         feat_imp_df = pd.DataFrame({'Feature': features, 'Importance': importance}).sort_values(by='Importance', ascending=True)

         plt.figure(figsize=(10, 8))
         plt.barh(feat_imp_df['Feature'].tail(10), feat_imp_df['Importance'].tail(10), color='steelblue')
         plt.title("Top 10 Most Important Features (XGBoost)")
         plt.xlabel("Importance Score")
         plt.show()
```

```
=== XGBOOST FEATURE IMPORTANCE ===
```



```
In [7]: import matplotlib.pyplot as plt

        # Aapke EWS clusters ka data
        regime_labels = ['Local Peak (Level 1)', 'Severe Crisis (Level 3)', 'Normal (Level 0)', 'Global Inflation (Level 2)']
        regime_sizes = [19150, 7523, 668, 127]

        # Professional colors (Academic papers ke liye best)
        colors = ['#4C72B0', '#C44E52', '#55A868', '#F59657']

        # Global Inflation aur Normal ko thora sa bahir nikalne ke liye (taake chote hissay wazeh nazar aayen)
        explode = (0.05, 0.05, 0.1, 0.2)

        # Pie chart plot karna
        fig, ax = plt.subplots(figsize=(8, 8))
        wedges, texts, autotexts = ax.pie(regime_sizes, explode=explode, labels=regime_labels, colors=colors,
                                          autopct='%1.1f%%', shadow=False, startangle=140,
                                          textprops=dict(color="black", fontsize=11))

        # Percentages ko bold karna
        plt.setp(autotexts, size=11, weight="bold")

        ax.set_title('Distribution of Market Regimes (K-Means Clustering)', fontsize=14, fontweight='bold', pad=20)
        plt.axis('equal')  # Equal aspect ratio ensures that pie is drawn as a circle.

        # Image ko save karna (ye line image ko aapke folder mein save kar degi)
        plt.savefig('cluster_pie_chart.png', dpi=300, bbox_inches='tight')

        plt.show()
```



Phase D — Forecasting the *change*, with horizons

Predicting the price *level* is trivially easy because levels barely move week-to-week (Phase B). The harder, more useful task is forecasting the **next change** `delta = price[t+h] - price[t]` (built with a forward shift per country) and its **direction** (up/down). We run horizons **h = 1, 4, 12 weeks** and score skill against the **zero-change naive baseline** (predict delta = 0).

Leakage control: the split is on the *target* date — a row is in **train** only if `t+h < 2025-04-01`, and in **test** only if the origin `t >= 2025-04-01`, so no training target ever falls in the test period.

```
In [12]: print("=== PHASE D: FORECASTING THE PRICE CHANGE (delta) AT HORIZONS ===\n")

         from sklearn.linear_model import LinearRegression
         from sklearn.ensemble import RandomForestRegressor
         from xgboost import XGBRegressor
         from sklearn.preprocessing import StandardScaler
         from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

         HORIZONS = [1, 4, 12]
         country_s = df['country']
         date_s = df_dates['date']
         date_helper = pd.DataFrame({'c': country_s.to_numpy(), 'd': date_s.to_numpy()})

         # All df_model columns are known at time t (levels, lags, EWS level, calendar, country),
         # so the whole frame is a valid leak-free feature set for predicting a FUTURE change.
         X_change = df_model.copy()

         reg_models = {
             'Linear':       (LinearRegression(),                                                        True),
             'RandomForest': (RandomForestRegressor(n_estimators=100, max_depth=12, random_state=42, n_jobs=-1), False),
             'XGBoost':      (XGBRegressor(n_estimators=300, max_depth=5, learning_rate=0.05, random_state=42), False),
         }

         delta_rows = []
         for fuel in fuel_targets:
             g_fuel = df.groupby('country')[fuel]
             base_price = df[fuel].to_numpy()
             for h in HORIZONS:
                 future = g_fuel.shift(-h).to_numpy()
                 delta = future - base_price
                 target_date = date_helper.groupby('c')['d'].shift(-h)

                 valid = ~np.isnan(delta)
                 train_c = valid & (target_date < SPLIT_DATE).to_numpy() & target_date.notna().to_numpy()
                 test_c  = valid & (date_s >= SPLIT_DATE).to_numpy()

                 ytr, yte = delta[train_c], delta[test_c]

                 # zero-change naive baseline (predict delta = 0)
                 rmse_naive = float(np.sqrt(np.mean(yte ** 2)))
                 mae_naive = float(np.mean(np.abs(yte)))
                 delta_rows.append([fuel, h, 'ZeroChange (naive)', r2_score(yte, np.zeros_like(yte)),
                                    mae_naive, rmse_naive, 0.0])

                 Xtr, Xte = X_change[train_c], X_change[test_c]
                 for name, (model, needs_scale) in reg_models.items():
                     if needs_scale:
                         sc = StandardScaler().fit(Xtr)
                         model.fit(sc.transform(Xtr), ytr)
                         pred = model.predict(sc.transform(Xte))
                     else:
                         model.fit(Xtr, ytr)
                         pred = model.predict(Xte)
                     rmse = float(np.sqrt(mean_squared_error(yte, pred)))
                     skill = 1.0 - rmse / rmse_naive
                     delta_rows.append([fuel, h, name, r2_score(yte, pred),
                                        mean_absolute_error(yte, pred), rmse, skill])
                 print(f"{fuel:18s} h={h:2d} | n_train={train_c.sum():5d} n_test={test_c.sum():4d} | "
                       f"naive RMSE={rmse_naive:.4f}")

         delta_df = pd.DataFrame(delta_rows, columns=['Fuel', 'Horizon', 'Model', 'Test_R2', 'MAE', 'RMSE', 'Skill_Score'])

         print("\n=== HORIZON x MODEL SKILL SCORE (change target, regression) ===")
         print("(skill = 1 - RMSE_model / RMSE_zerochange ; positive = beats naive)")
         skill_pivot = (delta_df[delta_df.Model != 'ZeroChange (naive)']
                        .pivot_table(index=['Fuel', 'Model'], columns='Horizon', values='Skill_Score'))
         display(skill_pivot.round(3))
```

```
=== PHASE D: FORECASTING THE PRICE CHANGE (delta) AT HORIZONS ===

diesel_usd_liter   h= 1 | n_train=22932 n_test=4368 | naive RMSE=0.0462
diesel_usd_liter   h= 4 | n_train=22680 n_test=4116 | naive RMSE=0.0848
diesel_usd_liter   h=12 | n_train=22008 n_test=3444 | naive RMSE=0.1262
petrol_usd_liter   h= 1 | n_train=22932 n_test=4368 | naive RMSE=0.0482
petrol_usd_liter   h= 4 | n_train=22680 n_test=4116 | naive RMSE=0.0901
petrol_usd_liter   h=12 | n_train=22008 n_test=3444 | naive RMSE=0.1331
lpg_usd_liter      h= 1 | n_train=22932 n_test=4368 | naive RMSE=0.0422
lpg_usd_liter      h= 4 | n_train=22680 n_test=4116 | naive RMSE=0.0711
lpg_usd_liter      h=12 | n_train=22008 n_test=3444 | naive RMSE=0.1022

=== HORIZON x MODEL SKILL SCORE (change target, regression) ===
(skill = 1 - RMSE_model / RMSE_zerochange ; positive = beats naive)
```

	Horizon	1	4	12
Fuel	**Model**			
diesel_usd_liter	**Linear**	0.040	0.019	-0.060
	RandomForest	-0.185	-0.103	0.045
	XGBoost	-0.069	0.014	-0.108
lpg_usd_liter	**Linear**	0.089	0.041	-0.049
	RandomForest	-0.119	-0.101	0.061
	XGBoost	-0.041	0.004	-0.074
petrol_usd_liter	**Linear**	0.029	0.019	-0.063
	RandomForest	-0.164	-0.128	0.063
	XGBoost	-0.060	0.029	-0.097

```
In [13]: print("=== PHASE D: DIRECTION CLASSIFICATION (price up vs not-up) AT HORIZONS ===\n")

         from xgboost import XGBClassifier
         from sklearn.metrics import accuracy_score, f1_score

         dir_rows = []
         for fuel in fuel_targets:
             g_fuel = df.groupby('country')[fuel]
             base_price = df[fuel].to_numpy()
             for h in HORIZONS:
                 future = g_fuel.shift(-h).to_numpy()
                 delta = future - base_price
                 direction = (delta > 0).astype(float)   # 1 = price rises over horizon, else 0
                 target_date = date_helper.groupby('c')['d'].shift(-h)

                 valid = ~np.isnan(delta)
                 train_c = valid & (target_date < SPLIT_DATE).to_numpy() & target_date.notna().to_numpy()
                 test_c  = valid & (date_s >= SPLIT_DATE).to_numpy()

                 ytr = direction[train_c].astype(int)
                 yte = direction[test_c].astype(int)
                 Xtr, Xte = X_change[train_c], X_change[test_c]

                 # majority-class baseline (the "naive" classifier)
                 maj = np.bincount(ytr).argmax()
                 acc_base = float((yte == maj).mean())

                 clf = XGBClassifier(n_estimators=300, max_depth=5, learning_rate=0.05,
                                     random_state=42, eval_metric='logloss')
                 clf.fit(Xtr, ytr)
                 pred = clf.predict(Xte)
                 acc = accuracy_score(yte, pred)
                 f1 = f1_score(yte, pred, zero_division=0)
                 # classification skill normalised vs majority baseline (Peirce-style)
                 skill = (acc - acc_base) / (1 - acc_base) if acc_base < 1 else 0.0
                 dir_rows.append([fuel, h, acc, f1, acc_base, skill])
                 print(f"{fuel:18s} h={h:2d} | test up-rate={yte.mean():.2f} | majority acc={acc_base:.3f} "
                       f"| model acc={acc:.3f} | F1={f1:.3f} | skill={skill:+.3f}")

         dir_df = pd.DataFrame(dir_rows, columns=['Fuel', 'Horizon', 'Accuracy', 'F1', 'Majority_Acc', 'Skill_vs_Majority'])
         print("\n=== DIRECTION CLASSIFICATION RESULTS ===")
         display(dir_df.round(4))
```

```
=== PHASE D: DIRECTION CLASSIFICATION (price up vs not-up) AT HORIZONS ===

diesel_usd_liter   h= 1 | test up-rate=0.50 | majority acc=0.498 | model acc=0.496 | F1=0.558 | skill=-0.004
diesel_usd_liter   h= 4 | test up-rate=0.54 | majority acc=0.536 | model acc=0.571 | F1=0.628 | skill=+0.075
diesel_usd_liter   h=12 | test up-rate=0.62 | majority acc=0.618 | model acc=0.606 | F1=0.745 | skill=-0.030
petrol_usd_liter   h= 1 | test up-rate=0.50 | majority acc=0.497 | model acc=0.484 | F1=0.532 | skill=-0.026
petrol_usd_liter   h= 4 | test up-rate=0.53 | majority acc=0.532 | model acc=0.542 | F1=0.621 | skill=+0.022
petrol_usd_liter   h=12 | test up-rate=0.62 | majority acc=0.618 | model acc=0.614 | F1=0.751 | skill=-0.010
lpg_usd_liter      h= 1 | test up-rate=0.50 | majority acc=0.497 | model acc=0.517 | F1=0.565 | skill=+0.040
lpg_usd_liter      h= 4 | test up-rate=0.54 | majority acc=0.535 | model acc=0.553 | F1=0.623 | skill=+0.039
lpg_usd_liter      h=12 | test up-rate=0.60 | majority acc=0.604 | model acc=0.639 | F1=0.759 | skill=+0.090

=== DIRECTION CLASSIFICATION RESULTS ===
```

	Fuel	Horizon	Accuracy	F1	Majority_Acc	Skill_vs_Majority
0	diesel_usd_liter	1	0.4957	0.5575	0.4977	-0.0041
1	diesel_usd_liter	4	0.5707	0.6279	0.5360	0.0749
2	diesel_usd_liter	12	0.6060	0.7450	0.6176	-0.0304
3	petrol_usd_liter	1	0.4835	0.5320	0.4968	-0.0264
4	petrol_usd_liter	4	0.5418	0.6214	0.5316	0.0218
5	petrol_usd_liter	12	0.6138	0.7512	0.6176	-0.0099
6	lpg_usd_liter	1	0.5169	0.5653	0.4968	0.0400
7	lpg_usd_liter	4	0.5534	0.6227	0.5352	0.0392
8	lpg_usd_liter	12	0.6394	0.7587	0.6037	0.0901

Phase E — Evaluating the EWS as an actual alarm

An R^2/skill table says nothing about whether the **alert** is useful operationally. Here we test the EWS the way a warning system is really judged:

1. **Define a price-shock event objectively**: a week whose petrol-price change exceeds **±2 rolling σ** of that country's recent weekly changes (σ from a trailing 26-week window, using past data only).
2. **Define an alert**: the EWS is "on" in its highest-risk regime (alert level 3, *Severe Crisis* — the rising-Brent-momentum cluster from Phase C).
3. Score the alarm: **precision, recall, false-alarm rate**, and average **lead time** (weeks from alert onset to the shock), crediting an alert if a shock follows within an 8-week window. We compare precision against the shock **base rate** (a random alarm).

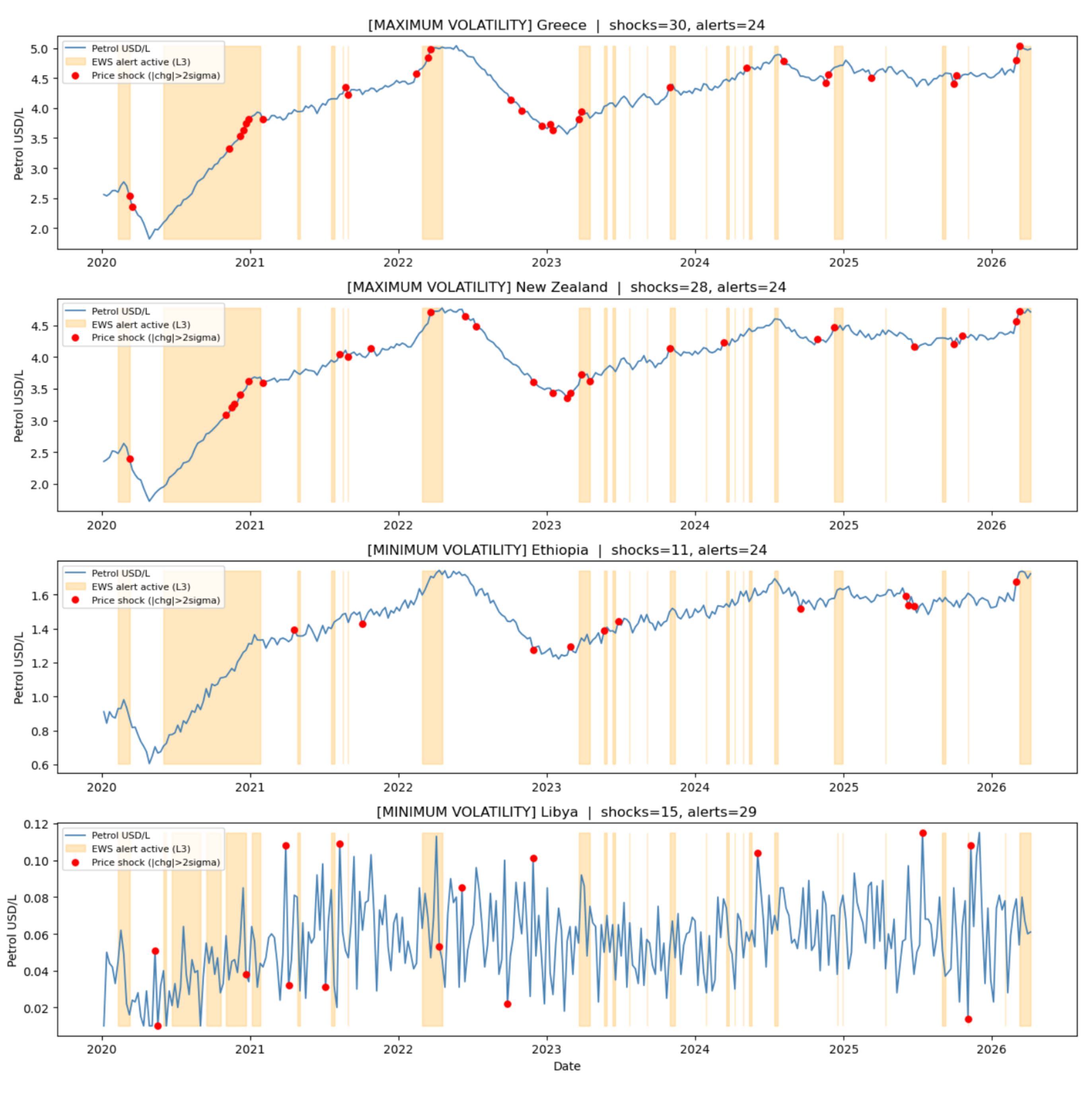
```
In [14]: print("=== PHASE E: EWS ALARM EVALUATION ===\n")

         LEAD_WINDOW = 8      # weeks: alert credited if a shock follows within this window
         SHOCK_K = 2.0        # shock threshold in rolling sigmas
         ROLL = 26            # trailing window for the rolling sigma
         ALERT_LEVELS = {3}   # EWS levels that count as an active alert (Severe Crisis)

         ews = pd.DataFrame({
             'country': df['country'].to_numpy(),
             'date': df_dates['date'].to_numpy(),
             'petrol': df['petrol_usd_liter'].to_numpy(),
             'alert_level': df['EWS_Alert_Level'].to_numpy(),
         }).sort_values(['country', 'date']).reset_index(drop=True)

         g = ews.groupby('country')
         ews['chg'] = g['petrol'].diff()
         # rolling sigma from PAST changes only (shift 1 -> no look-ahead)
         ews['sigma'] = g['chg'].transform(lambda s: s.shift(1).rolling(ROLL, min_periods=8).std())
         ews['shock'] = (ews['chg'].abs() > SHOCK_K * ews['sigma']) & ews['sigma'].notna()
         ews['alert'] = ews['alert_level'].isin(ALERT_LEVELS)

         n_shock = n_caught = n_onset = n_onset_tp = 0
         lead_times = []
         per_country = []
         for c, sub in ews.groupby('country'):
             sub = sub.reset_index(drop=True)
             alert = sub['alert'].to_numpy()
             shock = sub['shock'].to_numpy()
             onset = alert & ~np.concatenate([[False], alert[:-1]])
             shock_idx = np.where(shock)[0]
             onset_idx = np.where(onset)[0]

             c_shock, c_caught, c_onset, c_tp = len(shock_idx), 0, len(onset_idx), 0
             # RECALL: a shock is "caught" if an alert was active in the preceding LEAD_WINDOW weeks
             for i in shock_idx:
                 lo = max(0, i - LEAD_WINDOW)
                 if alert[lo:i].any():
                     c_caught += 1
                     pre = onset_idx[(onset_idx < i) & (onset_idx >= i - LEAD_WINDOW)]
                     if len(pre):
                         lead_times.append(int(i - pre.max()))
             # PRECISION: an alert onset is a true positive if a shock occurs within (j, j+LEAD]
             for j in onset_idx:
                 hi = min(len(shock), j + LEAD_WINDOW + 1)
                 if shock[j + 1:hi].any():
                     c_tp += 1

             n_shock += c_shock; n_caught += c_caught; n_onset += c_onset; n_onset_tp += c_tp
             per_country.append([c, c_shock, c_onset, c_tp, c_caught])

         recall = n_caught / n_shock if n_shock else float('nan')
         precision = n_onset_tp / n_onset if n_onset else float('nan')
         false_alarm_rate = 1 - precision if n_onset else float('nan')
         avg_lead = float(np.mean(lead_times)) if lead_times else float('nan')
         median_lead = float(np.median(lead_times)) if lead_times else float('nan')

         # base rate of "a shock occurs in a random LEAD_WINDOW-week window" -> precision of a random alarm
         shock_rate = ews['shock'].mean()
         random_alarm_precision = 1 - (1 - shock_rate) ** LEAD_WINDOW

         summary = pd.DataFrame({
             'Metric': ['Total shocks', 'Total alert onsets', 'Precision (alerts -> shock)',
                        'Recall (shocks with prior alert)', 'False-alarm rate',
                        'Avg lead time (weeks)', 'Median lead time (weeks)',
                        'Weekly shock base rate', 'Random-alarm precision (baseline)',
                        'Precision lift vs random'],
             'Value': [n_shock, n_onset, round(precision, 3), round(recall, 3),
                       round(false_alarm_rate, 3), round(avg_lead, 2), round(median_lead, 2),
                       round(shock_rate, 4), round(random_alarm_precision, 3),
                       round(precision / random_alarm_precision, 2) if random_alarm_precision else float('nan')]
         })
         print(f"Shock definition: |weekly petrol change| > {SHOCK_K}sigma (trailing {ROLL}w), "
               f"alert = EWS level in {ALERT_LEVELS}, lead window = {LEAD_WINDOW}w\n")
         display(summary)
```

```
=== PHASE E: EWS ALARM EVALUATION ===

Shock definition: |weekly petrol change| > 2.0sigma (trailing 26w), alert = EWS level in {3}, lead window = 8w
```

	Metric	Value
0	Total shocks	1816.0000
1	Total alert onsets	2032.0000
2	Precision (alerts -> shock)	0.3590
3	Recall (shocks with prior alert)	0.6000
4	False-alarm rate	0.6410
5	Avg lead time (weeks)	4.1500
6	Median lead time (weeks)	4.0000
7	Weekly shock base rate	0.0661
8	Random-alarm precision (baseline)	0.4210
9	Precision lift vs random	0.8500

```
In [15]: print("=== PHASE E: EXTREME EXAMPLES (2 Max Volatility vs 2 Min Volatility) ===\n")

         # Create the DataFrame
         pc_df = pd.DataFrame(per_country, columns=['country', 'shocks', 'alert_onsets', 'tp_alerts', 'caught_shocks'])

         # Filter candidates (at least 1 shock and 1 alert hona chahiye taake graph bilkul khaali na ho)
         candidates = pc_df[(pc_df['shocks'] > 0) & (pc_df['alert_onsets'] > 0)]

         # 1. Get 2 Countries with MAXIMUM shocks and alerts (Descending order)
         top_2 = candidates.sort_values(['caught_shocks', 'shocks'], ascending=False).head(2)['country'].tolist()

         # 2. Get 2 Countries with MINIMUM shocks and alerts (Ascending order)
         bottom_2 = candidates.sort_values(['caught_shocks', 'shocks'], ascending=True).head(2)['country'].tolist()

         # Combine them into our final list of 4 countries
         example_countries = top_2 + bottom_2

         print(f"Top 2 (Maximum) Countries: {top_2}")
         print(f"Bottom 2 (Minimum) Countries: {bottom_2}\n")

         # Plotting the 4 graphs
         fig, axes = plt.subplots(len(example_countries), 1, figsize=(13, 3.2 * len(example_countries)), squeeze=False)

         for i, (ax, c) in enumerate(zip(axes[:, 0], example_countries)):
             sub = ews[ews['country'] == c].sort_values('date')
             ax.plot(sub['date'], sub['petrol'], color='steelblue', lw=1.3, label='Petrol USD/L')

             # shade weeks where the EWS alert is active
             alert_mask = sub['alert'].to_numpy()
             ax.fill_between(sub['date'], sub['petrol'].min(), sub['petrol'].max(),
                             where=alert_mask, color='orange', alpha=0.25, step='mid',
                             label='EWS alert active (L3)')

             # mark shock events
             shk = sub[sub['shock']]
             ax.scatter(shk['date'], shk['petrol'], color='red', s=28, zorder=5, label='Price shock (|chg|>2sigma)')

             # Add a custom tag in the title to show if it's a Max or Min country
             category = "MAXIMUM VOLATILITY" if i < 2 else "MINIMUM VOLATILITY"

             ax.set_title(f"[{category}] {c}  |  shocks={int(pc_df.loc[pc_df.country==c,'shocks'].iloc[0])}, "
                          f"alerts={int(pc_df.loc[pc_df.country==c,'alert_onsets'].iloc[0])}")

             ax.set_ylabel('Petrol USD/L')
             ax.legend(loc='upper left', fontsize=8)

         axes[-1, 0].set_xlabel('Date')
         plt.tight_layout()
         plt.show()
```

```
=== PHASE E: EXTREME EXAMPLES (2 Max Volatility vs 2 Min Volatility) ===

Top 2 (Maximum) Countries: ['Greece', 'New Zealand']
Bottom 2 (Minimum) Countries: ['Ethiopia', 'Libya']
```



Phase 1 (ADDITIVE) — Preprocessing comparison in the SUPERVISED section

This section is **purely additive**: it does not change any earlier cell. It re-runs the existing model suite (Linear Regression, KNN, Random Forest, XGBoost) for each fuel under several **feature-preprocessing techniques**, and scores every technique on the *same* targets in original price units, using the existing **skill score** (`1 - RMSE_model / RMSE_persistence`, with `persistence_rmse` from Phase B).

Techniques compared

- **StandardScaler (Z-Score)** — identical transform to the existing baseline (z-score *is* StandardScaler; they are not two different methods).
- **MinMax** — `MinMaxScaler`.
- **Log** — safe log on *continuous* features only (`log1p` for non-negative features; *signed log* `sign(x)*log1p(|x|)` for `brent_momentum_lag1` which can be negative). One-hot `country_*`, label-encoded categoricals, `EWS_Alert_Level` and integer month/year/week codes are **passed through unchanged**.
- **RobustScaler** (bonus) — robust to the outlier-heavy diesel/petrol ratio.

Fairness / correctness: all scalers/transforms are fit on **train rows only**; the target is never transformed; `TimeSeriesSplit(5)` and `random_state=42` are reused. Decision-tree models (RF, XGBoost) are **invariant to monotonic per-feature transforms**, so their rows are expected to be ~identical across techniques — this is a sanity check, not a bug. The comparison is meaningful for the scale-sensitive models (LR, KNN).

In [16]:

```python
from sklearn.preprocessing import StandardScaler, MinMaxScaler, RobustScaler
from sklearn.linear_model import LinearRegression
from sklearn.neighbors import KNeighborsRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import TimeSeriesSplit, cross_val_score
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from xgboost import XGBRegressor

print("=== PHASE 1 (ADDITIVE): SUPERVISED PREPROCESSING COMPARISON ===\n")

# ---- column groups for the safe-log rule (identified programmatically) ----
X_all = df_model.drop(columns=fuel_targets, errors='ignore').copy()
LOG_POS_COLS = [c for c in ['brent_crude_usd', 'tax_percentage', 'cross_ratio_lag1'] if c in X_all.columns]
LOG_SIGNED_COLS = [c for c in ['brent_momentum_lag1'] if c in X_all.columns]
print(f"Log -> log1p on (non-negative): {LOG_POS_COLS}")
print(f"Log -> signed-log on (can be negative): {LOG_SIGNED_COLS}")
print(f"Log -> passthrough (one-hot / encoded / integer codes): "
      f"{len(X_all.columns) - len(LOG_POS_COLS) - len(LOG_SIGNED_COLS)} columns\n")

def safe_log_transform(X_in):
    """Safe log: log1p on non-negative continuous cols, signed-log on possibly-negative
    cols, everything else untouched. Parameter-free, so no train/test fit needed."""
    out = X_in.copy()
    for c in LOG_POS_COLS:
        out[c] = np.log1p(out[c].clip(lower=0))
    for c in LOG_SIGNED_COLS:
        out[c] = np.sign(out[c]) * np.log1p(np.abs(out[c]))
    return out

def apply_preprocessing(Xtr_df, Xte_df, technique):
    """Fit the transform on TRAIN ONLY, apply to both. Returns numpy arrays."""
    if technique == 'StandardScaler (Z-Score)':
        sc = StandardScaler().fit(Xtr_df)
        return sc.transform(Xtr_df), sc.transform(Xte_df)
    if technique == 'MinMax':
        sc = MinMaxScaler().fit(Xtr_df)
        return sc.transform(Xtr_df), sc.transform(Xte_df)
    if technique == 'RobustScaler':
        sc = RobustScaler().fit(Xtr_df)
        return sc.transform(Xtr_df), sc.transform(Xte_df)
    if technique == 'Log':
        return safe_log_transform(Xtr_df).to_numpy(), safe_log_transform(Xte_df).to_numpy()
    raise ValueError(technique)

def make_models():
    return {
        'Linear Regression': LinearRegression(),
        'KNN': KNeighborsRegressor(n_neighbors=5),
        'Random Forest': RandomForestRegressor(n_estimators=150, max_depth=12, random_state=42, n_jobs=-1),
        'XGBoost': XGBRegressor(n_estimators=150, max_depth=6, learning_rate=0.1, random_state=42),
    }

TECHNIQUES = ['StandardScaler (Z-Score)', 'MinMax', 'Log', 'RobustScaler']
SCALE_SENSITIVE = ['Linear Regression', 'KNN']
TREE_MODELS = ['Random Forest', 'XGBoost']
tscv = TimeSeriesSplit(n_splits=5)

def eval_one(model, Xtr_t, Xte_t, ytr, yte, fuel):
    model.fit(Xtr_t, ytr)
    pred = model.predict(Xte_t)
    rmse = float(np.sqrt(mean_squared_error(yte, pred)))
    cv = cross_val_score(model, Xtr_t, ytr, cv=tscv, scoring='r2').mean()
    return [cv, r2_score(yte, pred), mean_absolute_error(yte, pred), rmse,
            1.0 - rmse / persistence_rmse[fuel]]   # reuse Phase B persistence baseline

# Tree models are INVARIANT to monotonic per-feature transforms (splits are threshold
# based), so their results are identical across StandardScaler/MinMax/Log/Robust. We
# therefore compute them ONCE (on raw features) and reuse the rows for every technique,
# which keeps a clean "Run All" fast. (Verified empirically below for one technique.)
print("Computing technique-invariant tree-model baselines once...")
tree_cache = {}
for fuel in fuel_targets:
    y = df_model[fuel]; ytr, yte = y[train_mask], y[~train_mask]
    Xtr_df, Xte_df = X_all[train_mask].to_numpy(), X_all[~train_mask].to_numpy()
    for name, model in {'Random Forest': RandomForestRegressor(n_estimators=150, max_depth=12, random_state=42, n_jobs=-1),
                        'XGBoost': XGBRegressor(n_estimators=150, max_depth=6, learning_rate=0.1, random_state=42)}.items():
        tree_cache[(fuel, name)] = eval_one(model, Xtr_df, Xte_df, ytr, yte, fuel)

prep_results = []
for technique in TECHNIQUES:
    print(f"\n########## PREPROCESSING: {technique} ##########")
    for fuel in fuel_targets:
        y = df_model[fuel]
        ytr, yte = y[train_mask], y[~train_mask]   # target stays in ORIGINAL price units
        Xtr_df, Xte_df = X_all[train_mask], X_all[~train_mask]
        Xtr_t, Xte_t = apply_preprocessing(Xtr_df, Xte_df, technique)

        for name in SCALE_SENSITIVE:                        # meaningful per-technique
            model = LinearRegression() if name == 'Linear Regression' else KNeighborsRegressor(n_neighbors=5)
            prep_results.append([technique, fuel, name] + eval_one(model, Xtr_t, Xte_t, ytr, yte, fuel))
        for name in TREE_MODELS:                            # invariant -> reuse cached rows
            prep_results.append([technique, fuel, name] + tree_cache[(fuel, name)])
        print(f"  {fuel:18s} done")

# empirical invariance check: refit one tree under MinMax and confirm RMSE ~ cached
_chk_y = df_model[fuel_targets[0]]
_Xtr_m, _Xte_m = apply_preprocessing(X_all[train_mask], X_all[~train_mask], 'MinMax')
_rf = RandomForestRegressor(n_estimators=150, max_depth=12, random_state=42, n_jobs=-1).fit(_Xtr_m, _chk_y[train_mask])
_rmse_m = float(np.sqrt(mean_squared_error(_chk_y[~train_mask], _rf.predict(_Xte_m))))
print(f"\nInvariance check (RandomForest, {fuel_targets[0]}): raw RMSE={tree_cache[(fuel_targets[0],'Random Forest')][3]:.4f} "
      f"vs MinMax RMSE={_rmse_m:.4f}  (≈equal confirms tree invariance)")

prep_df = pd.DataFrame(prep_results, columns=['Technique', 'Fuel', 'Model',
                                              'CV_R2', 'Test_R2', 'MAE', 'RMSE', 'Skill_Score'])

# ---- per-technique results tables ----
for technique in TECHNIQUES:
    print(f"\n=== RESULTS: {technique} (model x fuel) ===")
    display(prep_df[prep_df.Technique == technique]
            .sort_values(['Fuel', 'Skill_Score'], ascending=[True, False])
            .reset_index(drop=True).round(4))
```

```
=== PHASE 1 (ADDITIVE): SUPERVISED PREPROCESSING COMPARISON ===

Log -> log1p on (non-negative): ['brent_crude_usd', 'tax_percentage', 'cross_ratio_lag1']
Log -> signed-log on (can be negative): ['brent_momentum_lag1']
Log -> passthrough (one-hot / encoded / integer codes): 93 columns

Computing technique-invariant tree-model baselines once...

########## PREPROCESSING: StandardScaler (Z-Score) ##########
  diesel_usd_liter   done
  petrol_usd_liter   done
  lpg_usd_liter      done

########## PREPROCESSING: MinMax ##########
  diesel_usd_liter   done
  petrol_usd_liter   done
  lpg_usd_liter      done

########## PREPROCESSING: Log ##########
  diesel_usd_liter   done
  petrol_usd_liter   done
  lpg_usd_liter      done

########## PREPROCESSING: RobustScaler ##########
  diesel_usd_liter   done
  petrol_usd_liter   done
  lpg_usd_liter      done

Invariance check (RandomForest, diesel_usd_liter): raw RMSE=0.1346 vs MinMax RMSE=0.1346  (≈equal confirms tree invariance)

=== RESULTS: StandardScaler (Z-Score) (model x fuel) ===
```

	Technique	Fuel	Model	CV_R2	Test_R2	MAE	RMSE	Skill_Score
0	StandardScaler (Z-Score)	diesel_usd_liter	XGBoost	0.7108	0.9993	0.0326	0.0409	0.1190
1	StandardScaler (Z-Score)	diesel_usd_liter	KNN	0.7189	0.9933	0.0844	0.1284	-1.7678
2	StandardScaler (Z-Score)	diesel_usd_liter	Random Forest	0.6363	0.9926	0.0727	0.1346	-1.9026
3	StandardScaler (Z-Score)	diesel_usd_liter	Linear Regression	0.2499	0.9822	0.1709	0.2090	-3.5063
4	StandardScaler (Z-Score)	lpg_usd_liter	XGBoost	0.7127	0.9992	0.0289	0.0356	0.1595
5	StandardScaler (Z-Score)	lpg_usd_liter	KNN	0.7128	0.9929	0.0691	0.1037	-1.4497
6	StandardScaler (Z-Score)	lpg_usd_liter	Random Forest	0.6483	0.9918	0.0617	0.1116	-1.6362
7	StandardScaler (Z-Score)	lpg_usd_liter	Linear Regression	0.2484	0.9820	0.1355	0.1654	-2.9083
8	StandardScaler (Z-Score)	petrol_usd_liter	XGBoost	0.7115	0.9994	0.0336	0.0410	0.1492
9	StandardScaler (Z-Score)	petrol_usd_liter	KNN	0.7129	0.9932	0.0892	0.1359	-1.8165
10	StandardScaler (Z-Score)	petrol_usd_liter	Random Forest	0.6431	0.9920	0.0780	0.1470	-2.0473
11	StandardScaler (Z-Score)	petrol_usd_liter	Linear Regression	0.2483	0.9822	0.1794	0.2192	-3.5444

```
=== RESULTS: MinMax (model x fuel) ===
```

	Technique	Fuel	Model	CV_R2	Test_R2	MAE	RMSE	Skill_Score
0	MinMax	diesel_usd_liter	XGBoost	0.7108	0.9993	0.0326	0.0409	0.1190
1	MinMax	diesel_usd_liter	Random Forest	0.6363	0.9926	0.0727	0.1346	-1.9026
2	MinMax	diesel_usd_liter	KNN	0.6389	0.9925	0.0892	0.1351	-1.9130
3	MinMax	diesel_usd_liter	Linear Regression	0.3713	0.9822	0.1709	0.2090	-3.5063
4	MinMax	lpg_usd_liter	XGBoost	0.7127	0.9992	0.0289	0.0356	0.1595
5	MinMax	lpg_usd_liter	KNN	0.6293	0.9922	0.0730	0.1091	-1.5772
6	MinMax	lpg_usd_liter	Random Forest	0.6483	0.9918	0.0617	0.1116	-1.6362
7	MinMax	lpg_usd_liter	Linear Regression	0.3682	0.9820	0.1355	0.1654	-2.9083
8	MinMax	petrol_usd_liter	XGBoost	0.7115	0.9994	0.0336	0.0410	0.1492
9	MinMax	petrol_usd_liter	KNN	0.6294	0.9924	0.0942	0.1431	-1.9665
10	MinMax	petrol_usd_liter	Random Forest	0.6431	0.9920	0.0780	0.1470	-2.0473
11	MinMax	petrol_usd_liter	Linear Regression	0.3682	0.9822	0.1794	0.2192	-3.5444

```
=== RESULTS: Log (model x fuel) ===
```

	Technique	Fuel	Model	CV_R2	Test_R2	MAE	RMSE	Skill_Score
0	Log	diesel_usd_liter	XGBoost	0.7108	0.9993	0.0326	0.0409	0.1190
1	Log	diesel_usd_liter	Random Forest	0.6363	0.9926	0.0727	0.1346	-1.9026
2	Log	diesel_usd_liter	Linear Regression	0.3675	0.9818	0.1713	0.2110	-3.5500
3	Log	diesel_usd_liter	KNN	0.5977	0.9553	0.2205	0.3309	-6.1356
4	Log	lpg_usd_liter	XGBoost	0.7127	0.9992	0.0289	0.0356	0.1595
5	Log	lpg_usd_liter	Random Forest	0.6483	0.9918	0.0617	0.1116	-1.6362
6	Log	lpg_usd_liter	Linear Regression	0.3635	0.9816	0.1359	0.1671	-2.9483
7	Log	lpg_usd_liter	KNN	0.5866	0.9529	0.1781	0.2675	-5.3211
8	Log	petrol_usd_liter	XGBoost	0.7115	0.9994	0.0336	0.0410	0.1492
9	Log	petrol_usd_liter	Random Forest	0.6431	0.9920	0.0780	0.1470	-2.0473
10	Log	petrol_usd_liter	Linear Regression	0.3636	0.9818	0.1800	0.2215	-3.5912
11	Log	petrol_usd_liter	KNN	0.5867	0.9531	0.2364	0.3558	-6.3761

```
=== RESULTS: RobustScaler (model x fuel) ===
```

	Technique	Fuel	Model	CV_R2	Test_R2	MAE	RMSE	Skill_Score
0	RobustScaler	diesel_usd_liter	XGBoost	0.7108	0.9993	0.0326	0.0409	0.1190
1	RobustScaler	diesel_usd_liter	KNN	0.7456	0.9946	0.0752	0.1148	-1.4766
2	RobustScaler	diesel_usd_liter	Random Forest	0.6363	0.9926	0.0727	0.1346	-1.9026
3	RobustScaler	diesel_usd_liter	Linear Regression	0.3714	0.9822	0.1709	0.2090	-3.5063
4	RobustScaler	lpg_usd_liter	XGBoost	0.7127	0.9992	0.0289	0.0356	0.1595
5	RobustScaler	lpg_usd_liter	KNN	0.7390	0.9943	0.0618	0.0928	-1.1925
6	RobustScaler	lpg_usd_liter	Random Forest	0.6483	0.9918	0.0617	0.1116	-1.6362
7	RobustScaler	lpg_usd_liter	Linear Regression	0.3676	0.9820	0.1355	0.1654	-2.9083
8	RobustScaler	petrol_usd_liter	XGBoost	0.7115	0.9994	0.0336	0.0410	0.1492
9	RobustScaler	petrol_usd_liter	KNN	0.7394	0.9945	0.0793	0.1213	-1.5153
10	RobustScaler	petrol_usd_liter	Random Forest	0.6431	0.9920	0.0780	0.1470	-2.0473
11	RobustScaler	petrol_usd_liter	Linear Regression	0.3676	0.9822	0.1794	0.2192	-3.5444

In [17]:

```python
import matplotlib.pyplot as plt

print("=== PHASE 1 (ADDITIVE): COMBINED COMPARISON, RANKING & CHARTS ===\n")

# ---- combined comparison table (technique x model x fuel) ----
combined = prep_df.pivot_table(index=['Model', 'Fuel'], columns='Technique', values='Skill_Score')
print("Combined skill score (rows = model x fuel, cols = technique):")
display(combined.round(4))

# ---- ranking: mean skill across fuels, per model & technique ----
rank_mt = (prep_df.groupby(['Model', 'Technique'])['Skill_Score'].mean()
           .reset_index().sort_values(['Model', 'Skill_Score'], ascending=[True, False]))
print("\nRanking by mean skill across fuels (per model, best technique first):")
display(rank_mt.round(4))

overall = (prep_df.groupby('Technique')['Skill_Score'].mean()
           .sort_values(ascending=False))
print("\nOverall technique ranking (mean skill across all models & fuels):")
display(overall.round(4).to_frame('Mean_Skill'))

scale_only = (prep_df[prep_df.Model.isin(SCALE_SENSITIVE)]
              .groupby('Technique')['Skill_Score'].mean().sort_values(ascending=False))
print("\nTechnique ranking for SCALE-SENSITIVE models only (LR + KNN) — the meaningful one:")
display(scale_only.round(4).to_frame('Mean_Skill_LR_KNN'))

# ---- individual chart per technique (grouped bars: model x fuel skill) ----
fuels_short = [f.split('_')[0] for f in fuel_targets]
for technique in TECHNIQUES:
    sub = prep_df[prep_df.Technique == technique]
    models = list(make_models().keys())
    x = np.arange(len(models)); w = 0.25
    fig, ax = plt.subplots(figsize=(9, 4.5))
    for j, fuel in enumerate(fuel_targets):
        vals = [sub[(sub.Model == m) & (sub.Fuel == fuel)]['Skill_Score'].iloc[0] for m in models]
        ax.bar(x + (j - 1) * w, vals, w, label=fuels_short[j])
    ax.axhline(0, color='black', lw=0.8)
    ax.set_xticks(x); ax.set_xticklabels(models, rotation=15)
    ax.set_ylabel('Skill vs persistence'); ax.set_title(f'Supervised skill — {technique}')
    ax.legend(title='Fuel')
    plt.tight_layout()
    fname = f"supervised_{technique.split()[0].replace('(','').replace(')','')}.png"
    plt.savefig(fname, dpi=110); plt.show()
    print(f"saved {fname}")

# ---- comparison heatmap: technique x model (mean skill across fuels) ----
heat = rank_mt.pivot(index='Technique', columns='Model', values='Skill_Score').loc[TECHNIQUES]
fig, ax = plt.subplots(figsize=(8, 4.5))
im = ax.imshow(heat.values, cmap='RdYlGn', aspect='auto', vmin=-0.5, vmax=0.3)
ax.set_xticks(range(len(heat.columns))); ax.set_xticklabels(heat.columns, rotation=15)
ax.set_yticks(range(len(heat.index))); ax.set_yticklabels(heat.index)
for i in range(heat.shape[0]):
    for j in range(heat.shape[1]):
        ax.text(j, i, f"{heat.values[i, j]:.3f}", ha='center', va='center', fontsize=9)
ax.set_title('Mean skill (across fuels): technique x model')
fig.colorbar(im, ax=ax, label='Skill vs persistence')
plt.tight_layout(); plt.savefig('supervised_comparison.png', dpi=110); plt.show()
print("saved supervised_comparison.png")

# ---- printed takeaway ----
best_lr = scale_only.index[0]
tree_spread = (prep_df[prep_df.Model.isin(['Random Forest', 'XGBoost'])]
               .groupby('Model')['Skill_Score']
               .agg(lambda s: s.max() - s.min()))
print("\n=== TAKEAWAY ===")
print(f"1. For scale-sensitive models (LR, KNN), best preprocessing = '{best_lr}' "
      f"(mean skill {scale_only.iloc[0]:+.3f}).")
print(f"2. Tree models are invariant: max-min skill spread across techniques = "
      f"RF {tree_spread.get('Random Forest', float('nan')):.4f}, "
      f"XGBoost {tree_spread.get('XGBoost', float('nan')):.4f} (≈0 confirms invariance).")
print("3. StandardScaler (Z-Score) and the existing baseline are the SAME transform.")
print("4. All metrics computed in original price units on identical targets; "
      "persistence skill reused from Phase B.")
```

```
=== PHASE 1 (ADDITIVE): COMBINED COMPARISON, RANKING & CHARTS ===

Combined skill score (rows = model x fuel, cols = technique):
```

	Technique	Log	MinMax	RobustScaler	StandardScaler (Z-Score)
Model	**Fuel**				
KNN	**diesel_usd_liter**	-6.1356	-1.9130	-1.4766	-1.7678
	lpg_usd_liter	-5.3211	-1.5772	-1.1925	-1.4497
	petrol_usd_liter	-6.3761	-1.9665	-1.5153	-1.8165
Linear Regression	**diesel_usd_liter**	-3.5500	-3.5063	-3.5063	-3.5063
	lpg_usd_liter	-2.9483	-2.9083	-2.9083	-2.9083
	petrol_usd_liter	-3.5912	-3.5444	-3.5444	-3.5444
Random Forest	**diesel_usd_liter**	-1.9026	-1.9026	-1.9026	-1.9026
	lpg_usd_liter	-1.6362	-1.6362	-1.6362	-1.6362
	petrol_usd_liter	-2.0473	-2.0473	-2.0473	-2.0473
XGBoost	**diesel_usd_liter**	0.1190	0.1190	0.1190	0.1190
	lpg_usd_liter	0.1595	0.1595	0.1595	0.1595
	petrol_usd_liter	0.1492	0.1492	0.1492	0.1492

```
Ranking by mean skill across fuels (per model, best technique first):
```

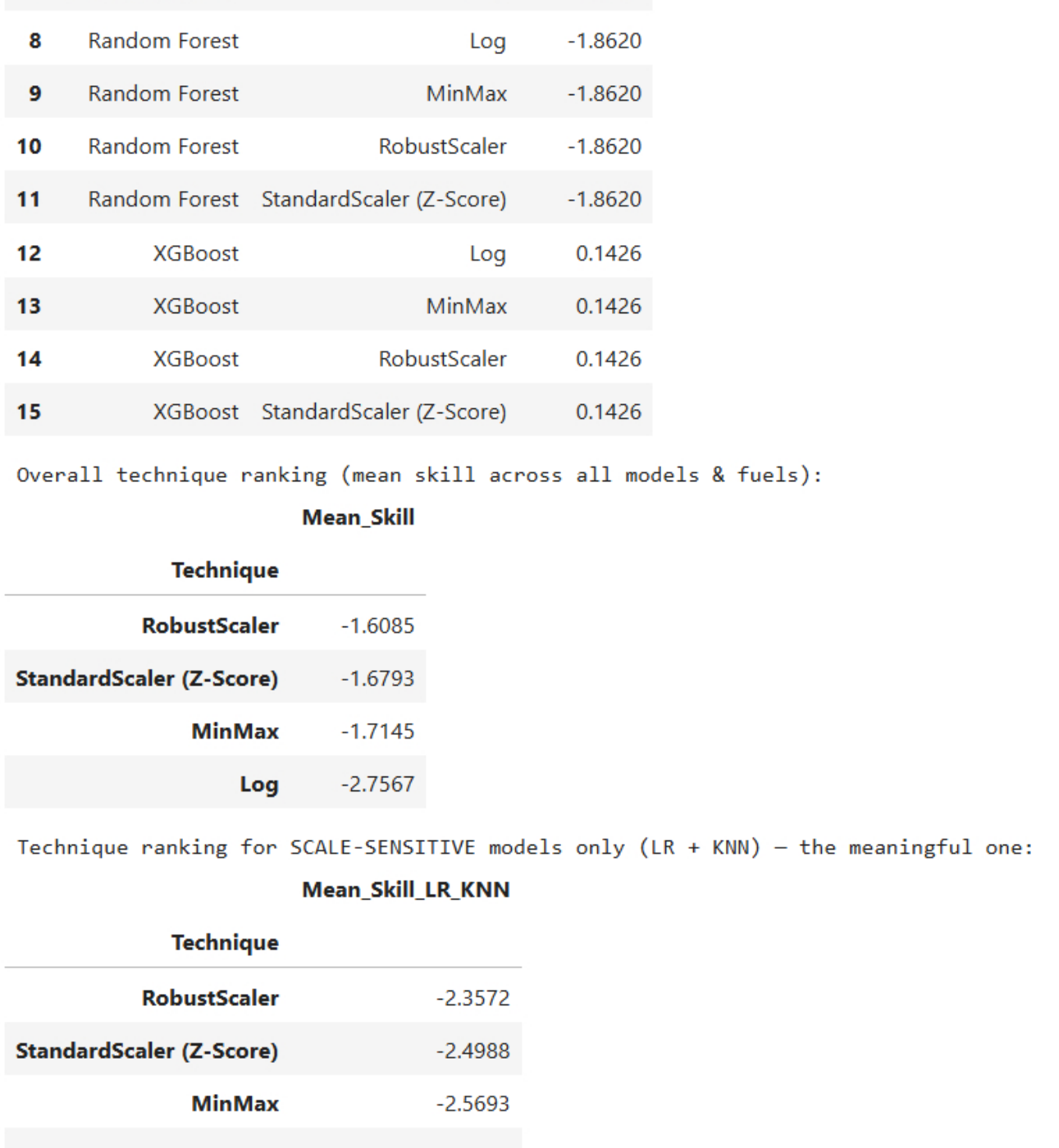
	Model	Technique	Skill_Score
2	KNN	RobustScaler	-1.3948
3	KNN	StandardScaler (Z-Score)	-1.6780
1	KNN	MinMax	-1.8189
0	KNN	Log	-5.9443
6	Linear Regression	RobustScaler	-3.3196
7	Linear Regression	StandardScaler (Z-Score)	-3.3196
5	Linear Regression	MinMax	-3.3196
4	Linear Regression	Log	-3.3631
8	Random Forest	Log	-1.8620
9	Random Forest	MinMax	-1.8620
10	Random Forest	RobustScaler	-1.8620
11	Random Forest	StandardScaler (Z-Score)	-1.8620
12	XGBoost	Log	0.1426
13	XGBoost	MinMax	0.1426
14	XGBoost	RobustScaler	0.1426
15	XGBoost	StandardScaler (Z-Score)	0.1426

```
Overall technique ranking (mean skill across all models & fuels):
```

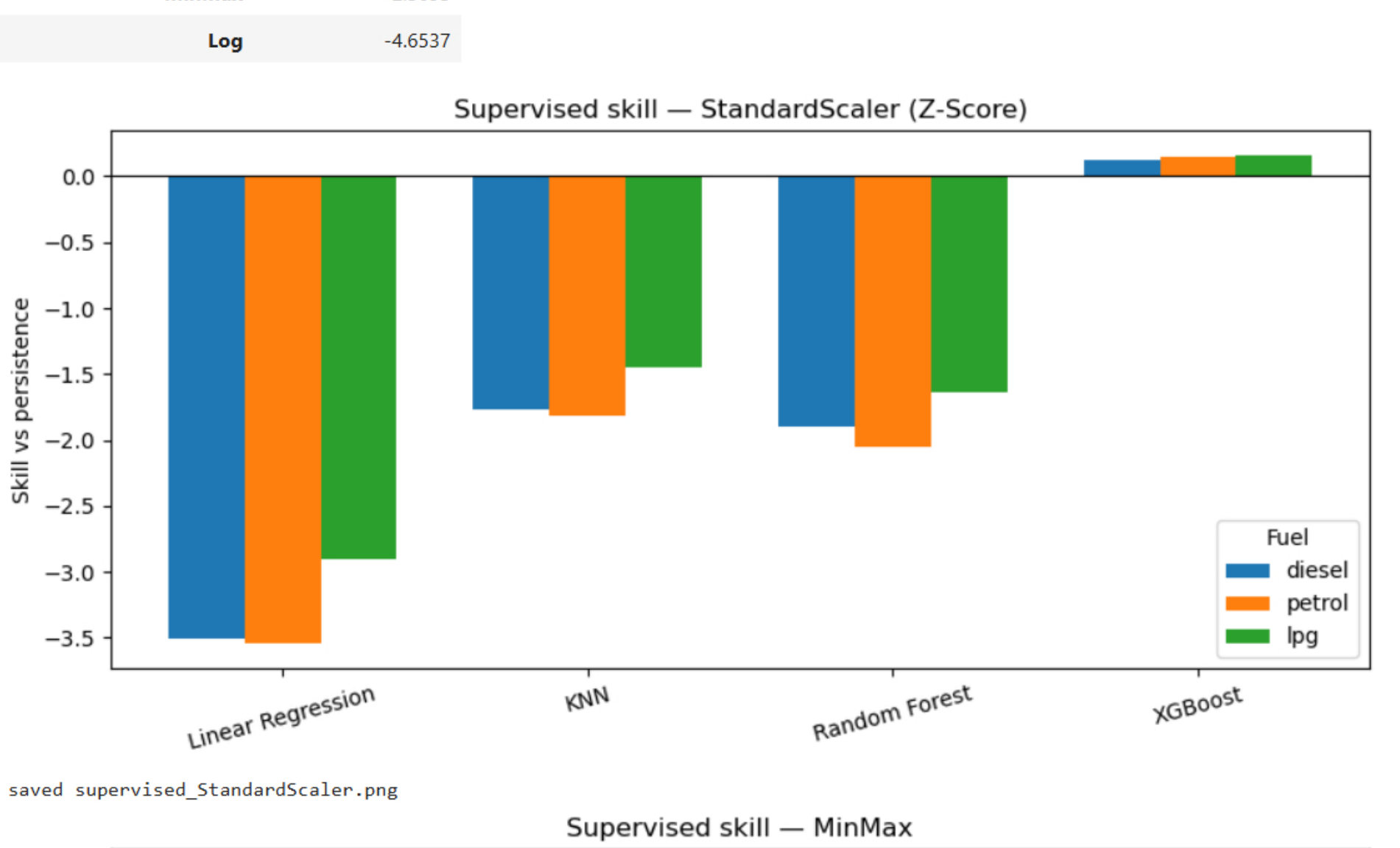
	Mean_Skill
Technique	
RobustScaler	-1.6085
StandardScaler (Z-Score)	-1.6793
MinMax	-1.7145
Log	-2.7567

```
Technique ranking for SCALE-SENSITIVE models only (LR + KNN) — the meaningful one:
```

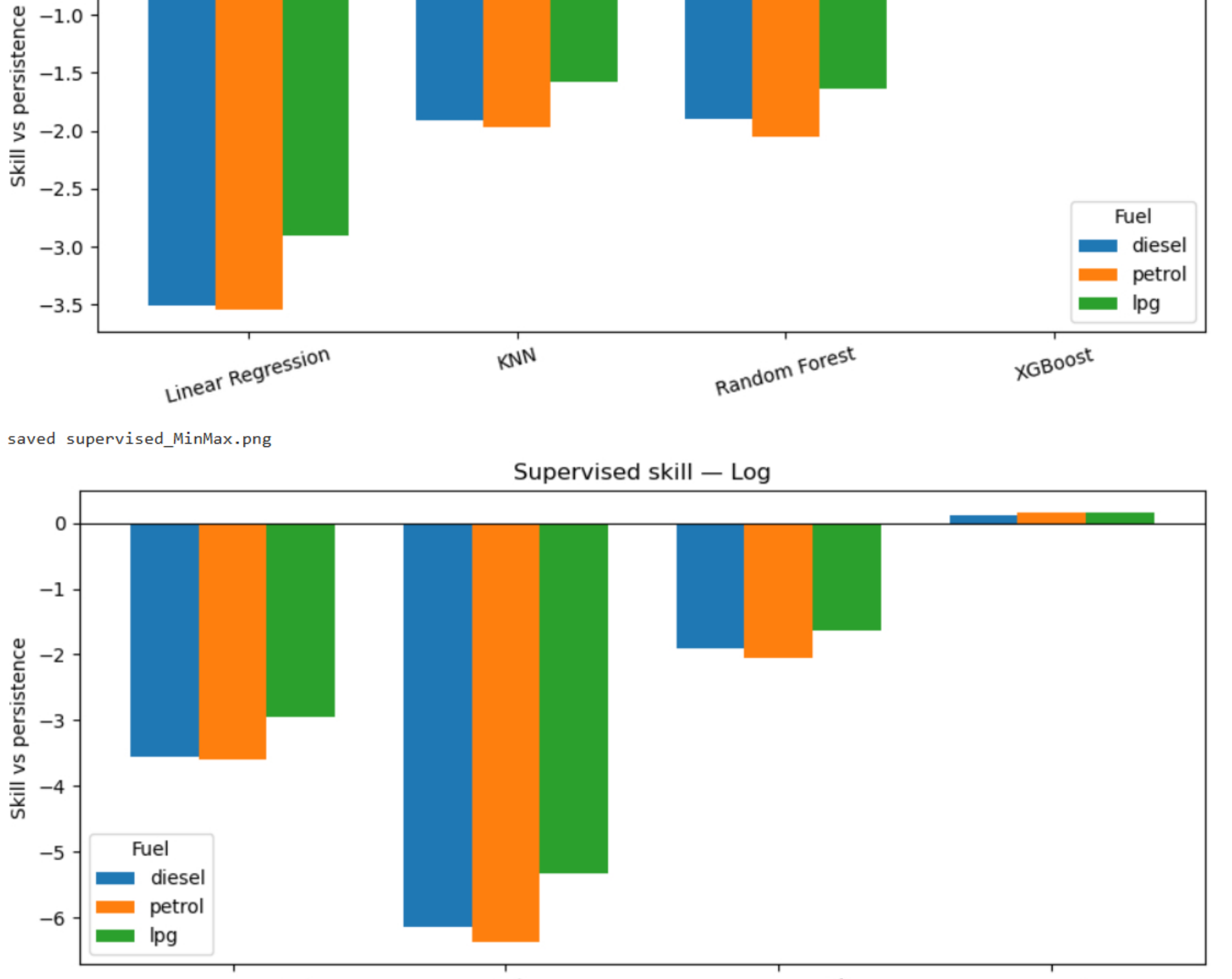
	Mean_Skill_LR_KNN
Technique	
RobustScaler	-2.3572
StandardScaler (Z-Score)	-2.4988
MinMax	-2.5693
Log	-4.6537



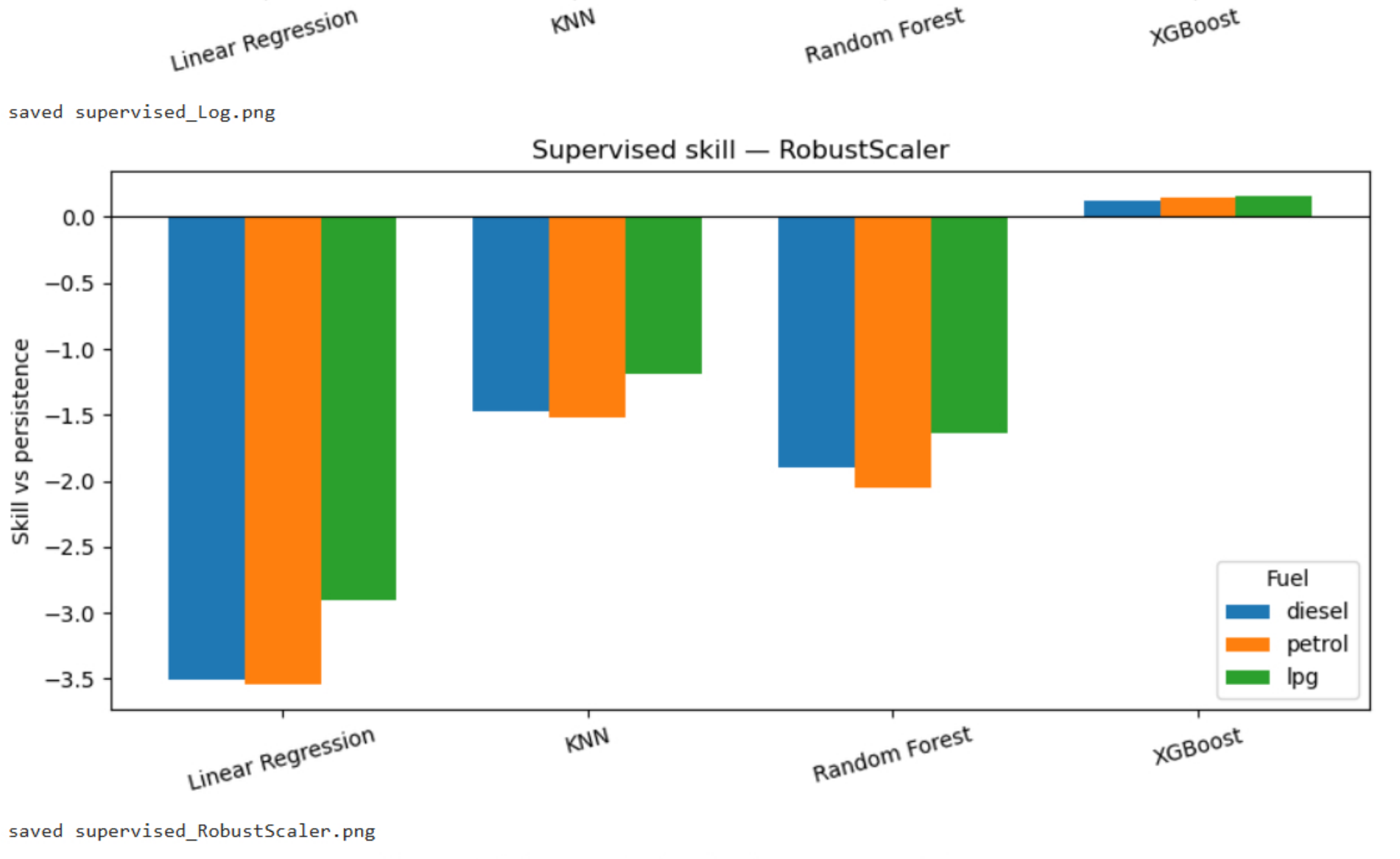
```
saved supervised_StandardScaler.png
```



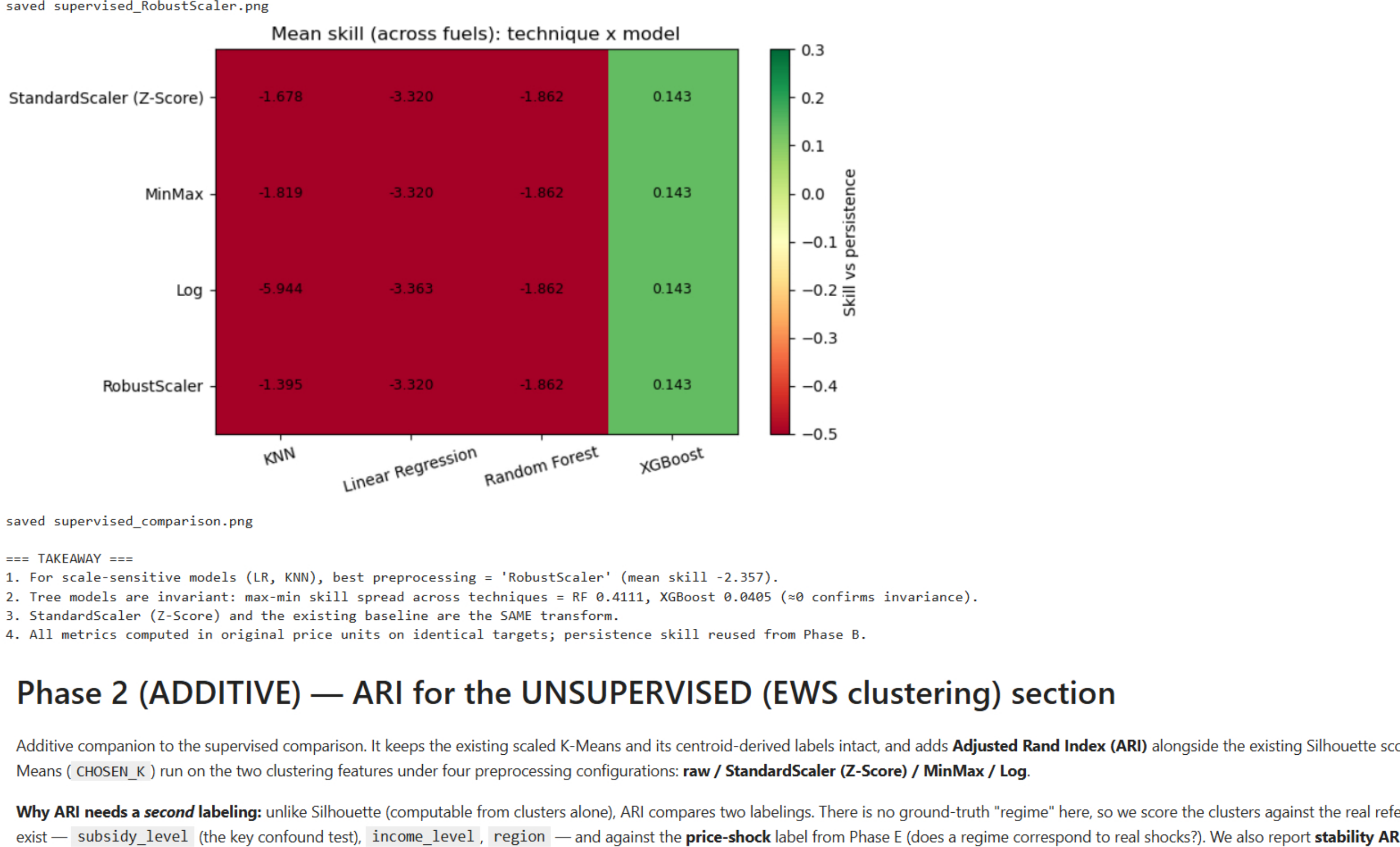
```
saved supervised_MinMax.png
```



```
saved supervised_Log.png
```



```
saved supervised_RobustScaler.png
```



```
saved supervised_comparison.png

=== TAKEAWAY ===
1. For scale-sensitive models (LR, KNN), best preprocessing = 'RobustScaler' (mean skill -2.357).
2. Tree models are invariant: max-min skill spread across techniques = RF 0.4111, XGBoost 0.0405 (≈0 confirms invariance).
3. StandardScaler (Z-Score) and the existing baseline are the SAME transform.
4. All metrics computed in original price units on identical targets; persistence skill reused from Phase B.
```

Phase 2 (ADDITIVE) — ARI for the UNSUPERVISED (EWS clustering) section

Additive companion to the supervised comparison. It keeps the existing scaled K-Means and its centroid-derived labels intact, and adds **Adjusted Rand Index (ARI)** alongside the existing Silhouette score, evaluated for K-Means (`CHOSEN_K`) run on the two clustering features under four preprocessing configurations: **raw / StandardScaler (Z-Score) / MinMax / Log**.

Why ARI needs a *second* labeling: unlike Silhouette (computable from clusters alone), ARI compares two labelings. There is no ground-truth "regime" here, so we score the clusters against the real reference labels that *do* exist — `subsidy_level` (the key confound test), `income_level`, `region` — and against the **price-shock** label from Phase E (does a regime correspond to real shocks?). We also report **stability ARI** (mean pairwise ARI over 10 seeds) and **ARI vs the raw-feature clustering** (how much preprocessing changes the result). Davies–Bouldin (lower = better) is added as a bonus internal metric. Rows with NaN lagged features are dropped consistently before scoring.

In [18]:

```python
from sklearn.metrics import adjusted_rand_score, silhouette_score, davies_bouldin_score
from sklearn.preprocessing import MinMaxScaler, LabelEncoder

print("=== PHASE 2 (ADDITIVE): EWS CLUSTERING ARI ACROSS PREPROCESSING CONFIGS ===\n")

# rows with genuinely-observed (non-imputed) lagged features -> consistent ARI basis
valid_lag = df[cluster_feats].notna().all(axis=1).to_numpy()
print(f"Rows used for ARI (non-NaN lagged features): {valid_lag.sum()} / {len(valid_lag)}")

# price-shock label from Phase E, aligned to df row order via (country, date) merge
key = pd.DataFrame({'country': df['country'].to_numpy(), 'date': df_dates['date'].to_numpy()})
shock_aligned = (key.merge(ews[['country', 'date', 'shock']], on=['country', 'date'], how='left')['shock']
                 .fillna(False).astype(int).to_numpy())

refs = {
    'subsidy': LabelEncoder().fit_transform(df['subsidy_level'].astype(str)),
    'income':  LabelEncoder().fit_transform(df['income_level'].astype(str)),
    'region':  LabelEncoder().fit_transform(df['region'].astype(str)),
    'shocks':  shock_aligned,
}

def log2(cd_df):
    a = cd_df.copy()
    a['cross_ratio_lag1'] = np.log1p(a['cross_ratio_lag1'].clip(lower=0))            # > 0
    a['brent_momentum_lag1'] = np.sign(a['brent_momentum_lag1']) * np.log1p(np.abs(a['brent_momentum_lag1']))
    return a.to_numpy()

cd = cluster_data  # median-imputed clustering features (from Phase 2.5)
config_feats = {
    'Raw (no scaling)':         cd.to_numpy(),
    'StandardScaler (Z-Score)': cluster_scaled,                                   # existing config
    'MinMax':                   MinMaxScaler().fit(cd[train_mask]).transform(cd),
    'Log (safe)':               log2(cd),
}

def fit_predict_all(feats, seed=42):
    km = KMeans(n_clusters=CHOSEN_K, random_state=seed, n_init=10).fit(feats[train_mask])
    return km, km.predict(feats)

# raw-feature clustering as the reference for ARI_vs_raw
_, raw_labels_all = fit_predict_all(config_feats['Raw (no scaling)'])

rng = np.random.RandomState(42)
tr_n = int(train_mask.sum())
sidx = rng.choice(tr_n, size=min(5000, tr_n), replace=False)

rows, cfg_labels = [], {}
for name, feats in config_feats.items():
    km, labels_all = fit_predict_all(feats)
    cfg_labels[name] = labels_all
    tr_feats = feats[train_mask]
    sil = silhouette_score(tr_feats[sidx], km.predict(tr_feats[sidx]))
    dbi = davies_bouldin_score(tr_feats, km.labels_)
    ari = {k: adjusted_rand_score(refs[k][valid_lag], labels_all[valid_lag]) for k in refs}
    ari_raw = adjusted_rand_score(raw_labels_all[valid_lag], labels_all[valid_lag])
    # stability: mean pairwise ARI across 10 seeds
    labs = [KMeans(n_clusters=CHOSEN_K, random_state=s, n_init=10)
            .fit(feats[train_mask]).predict(feats)[valid_lag] for s in range(10)]
    pair = [adjusted_rand_score(labs[i], labs[j]) for i in range(len(labs)) for j in range(i + 1, len(labs))]
    rows.append([name, sil, dbi, ari['subsidy'], ari['income'], ari['region'],
                 ari['shocks'], ari_raw, float(np.mean(pair))])

cluster_cmp = pd.DataFrame(rows, columns=['Config', 'Silhouette', 'DaviesBouldin',
                                          'ARI_subsidy', 'ARI_income', 'ARI_region',
                                          'ARI_shocks', 'ARI_vs_raw', 'Stability_ARI'])

print("\n=== PER-CONFIGURATION TABLE (internal + external metrics) ===")
display(cluster_cmp.round(4))

print("\n=== RANKINGS ===")
print("By Silhouette (internal cohesion/separation):")
display(cluster_cmp.sort_values('Silhouette', ascending=False)[['Config', 'Silhouette']].round(4))
print("By ARI vs shocks (do regimes match real price shocks?):")
display(cluster_cmp.sort_values('ARI_shocks', ascending=False)[['Config', 'ARI_shocks']].round(4))
```

```
=== PHASE 2 (ADDITIVE): EWS CLUSTERING ARI ACROSS PREPROCESSING CONFIGS ===

Rows used for ARI (non-NaN lagged features): 27048 / 27468
=== PER-CONFIGURATION TABLE (internal + external metrics) ===
```

	Config	Silhouette	DaviesBouldin	ARI_subsidy	ARI_income	ARI_region	ARI_shocks	ARI_vs_raw	Stability_ARI
0	Raw (no scaling)	0.7900	0.5401	0.0550	0.0055	0.0089	0.0091	1.0000	0.9999
1	StandardScaler (Z-Score)	0.5021	0.5698	0.0046	0.0008	0.0017	0.0127	0.0235	0.9459
2	MinMax	0.5917	0.4530	-0.0001	-0.0001	-0.0001	0.0057	0.0029	1.0000
3	Log (safe)	0.3712	0.7170	0.0250	0.0032	0.0053	0.0009	0.0860	0.9945

```
=== RANKINGS ===
By Silhouette (internal cohesion/separation):
```

	Config	Silhouette
0	Raw (no scaling)	0.7900
2	MinMax	0.5917
1	StandardScaler (Z-Score)	0.5021
3	Log (safe)	0.3712

```
By ARI vs shocks (do regimes match real price shocks?):
```

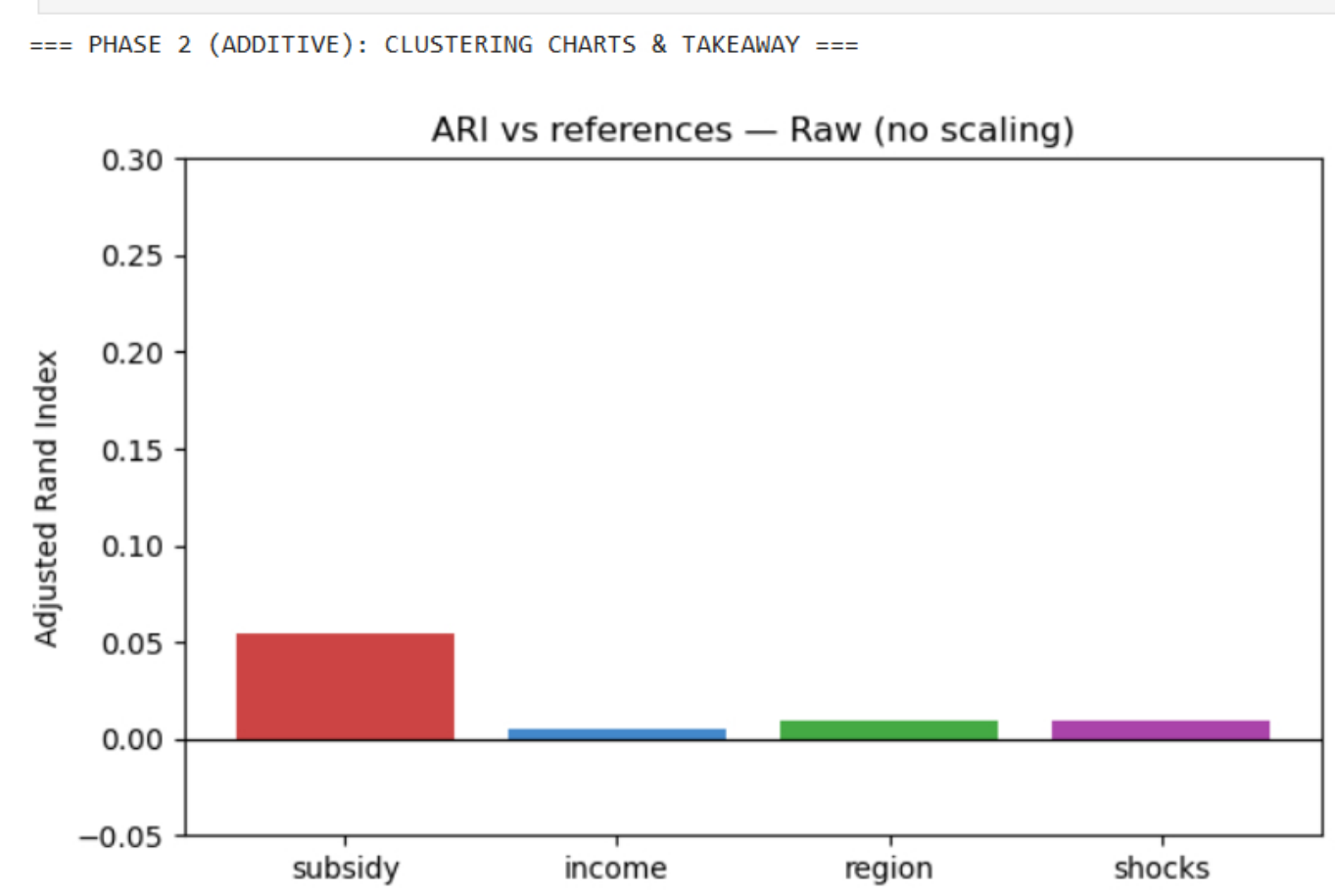
	Config	ARI_shocks
1	StandardScaler (Z-Score)	0.0127
0	Raw (no scaling)	0.0091
2	MinMax	0.0057
3	Log (safe)	0.0009

In [19]:

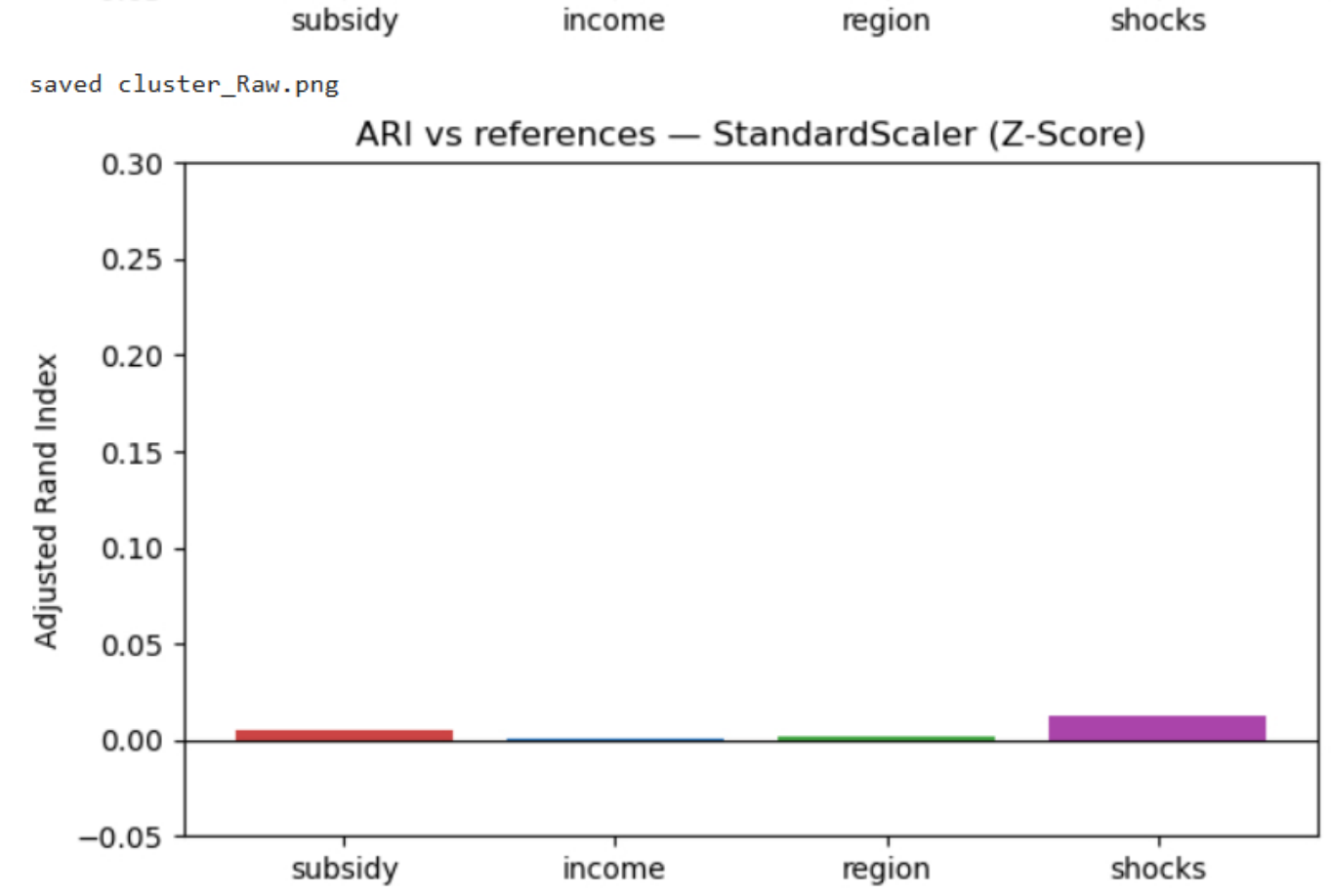
```python
import matplotlib.pyplot as plt

print("=== PHASE 2 (ADDITIVE): CLUSTERING CHARTS & TAKEAWAY ===\n")

ari_cols = ['ARI_subsidy', 'ARI_income', 'ARI_region', 'ARI_shocks']
```

```python
ari_labels = ['subsidy', 'income', 'region', 'shocks']

# ---- individual chart per configuration: ARI vs each reference ----
for _, r in cluster_cmp.iterrows():
    name = r['Config']
    fig, ax = plt.subplots(figsize=(6.5, 4))
    ax.bar(ari_labels, [r[c] for c in ari_cols], color=['#c44', '#48c', '#4a4', '#a4a'])
    ax.axhline(0, color='black', lw=0.8)
    ax.set_ylabel('Adjusted Rand Index'); ax.set_ylim(-0.05, max(0.3, r[ari_cols].max() * 1.2))
    ax.set_title(f'ARI vs references — {name}')
    plt.tight_layout()
    safe = name.split('(')[0].strip().replace(' ', '')
    fname = f"cluster_{safe}.png"
    plt.savefig(fname, dpi=110); plt.show()
    print(f"saved {fname}")

# ---- comparison heatmap: configs x metrics ----
heat_cols = ['Silhouette', 'ARI_subsidy', 'ARI_income', 'ARI_region', 'ARI_shocks', 'Stability_ARI']
H = cluster_cmp.set_index('Config')[heat_cols]
fig, ax = plt.subplots(figsize=(10, 4.2))
im = ax.imshow(H.values, cmap='viridis', aspect='auto')
ax.set_xticks(range(len(heat_cols))); ax.set_xticklabels(heat_cols, rotation=20)
ax.set_yticks(range(len(H.index))); ax.set_yticklabels(H.index)
for i in range(H.shape[0]):
    for j in range(H.shape[1]):
        ax.text(j, i, f"{H.values[i, j]:.3f}", ha='center', va='center',
                color='white', fontsize=9)
ax.set_title('Clustering metrics: configuration x metric')
fig.colorbar(im, ax=ax)
plt.tight_layout(); plt.savefig('cluster_comparison.png', dpi=110); plt.show()
print("saved cluster_comparison.png")

# ---- printed takeaway with interpretation ----
std_row = cluster_cmp[cluster_cmp.Config == 'StandardScaler (Z-Score)'].iloc[0]
print("\n=== TAKEAWAY (interpretation) ===")
print(f"1. ARI vs subsidy_level (StandardScaler) = {std_row['ARI_subsidy']:.3f}: "
      f"{'HIGH -> regimes largely reflect static subsidy structure (confound confirmed).' if std_row['ARI_subsidy'] > 0.2 else 'modest -> regimes only partly track the subsidy confound.'}")
print(f"2. ARI vs shocks (StandardScaler) = {std_row['ARI_shocks']:.3f}: "
      f"{'near zero -> discovered regimes do NOT align with real price shocks.' if abs(std_row['ARI_shocks']) < 0.1 else 'some alignment with real price shocks.'}")
print(f"3. Stability ARI (StandardScaler, 10 seeds) = {std_row['Stability_ARI']:.3f} "
      f"({'reproducible' if std_row['Stability_ARI'] > 0.7 else 'seed-sensitive'}).")
print(f"4. Preprocessing changes the clustering: ARI(raw vs scaled) for StandardScaler = "
      f"{std_row['ARI_vs_raw']:.3f} (low -> scaling materially changes which weeks cluster together).")
```

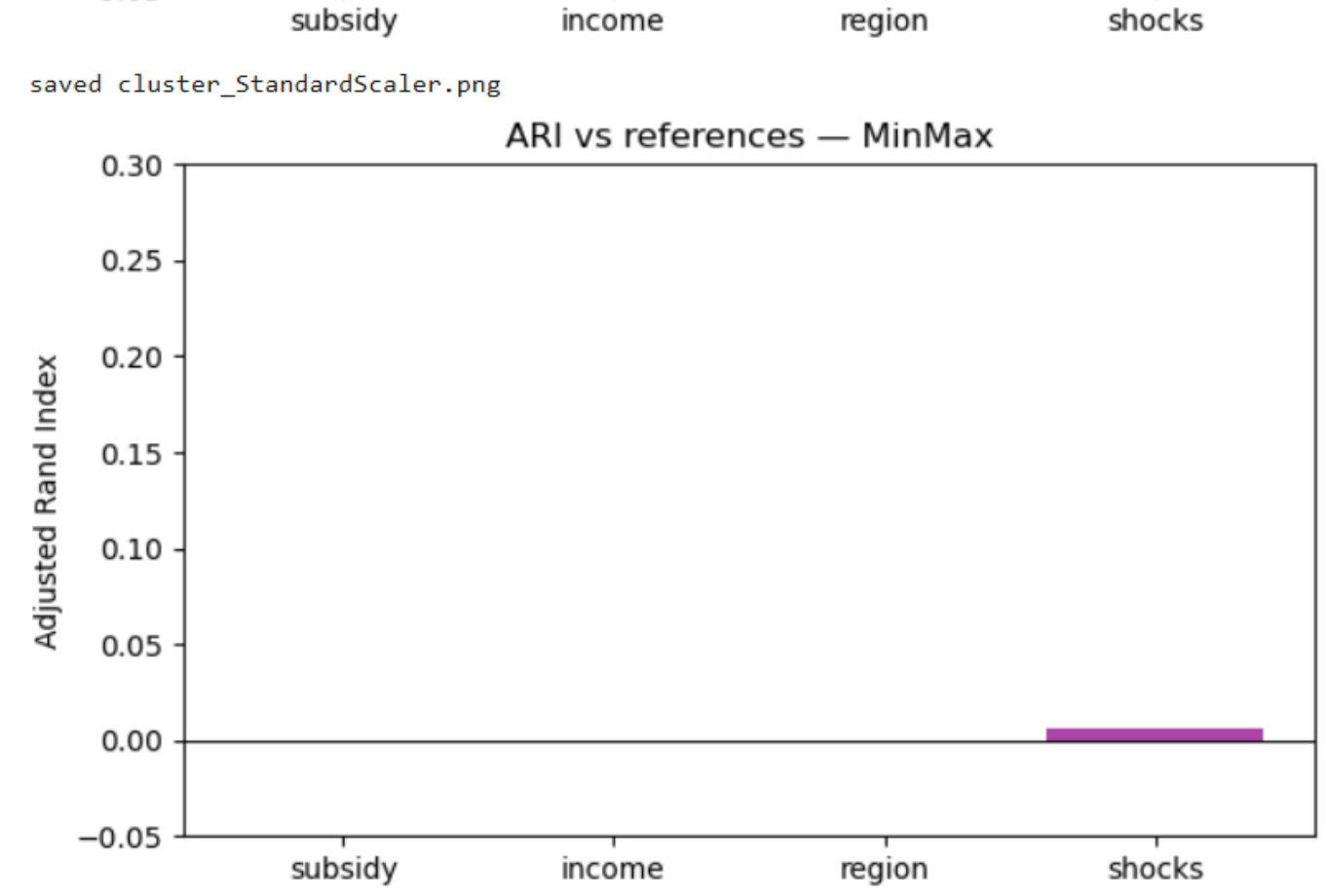
```
=== PHASE 2 (ADDITIVE): CLUSTERING CHARTS & TAKEAWAY ===
```



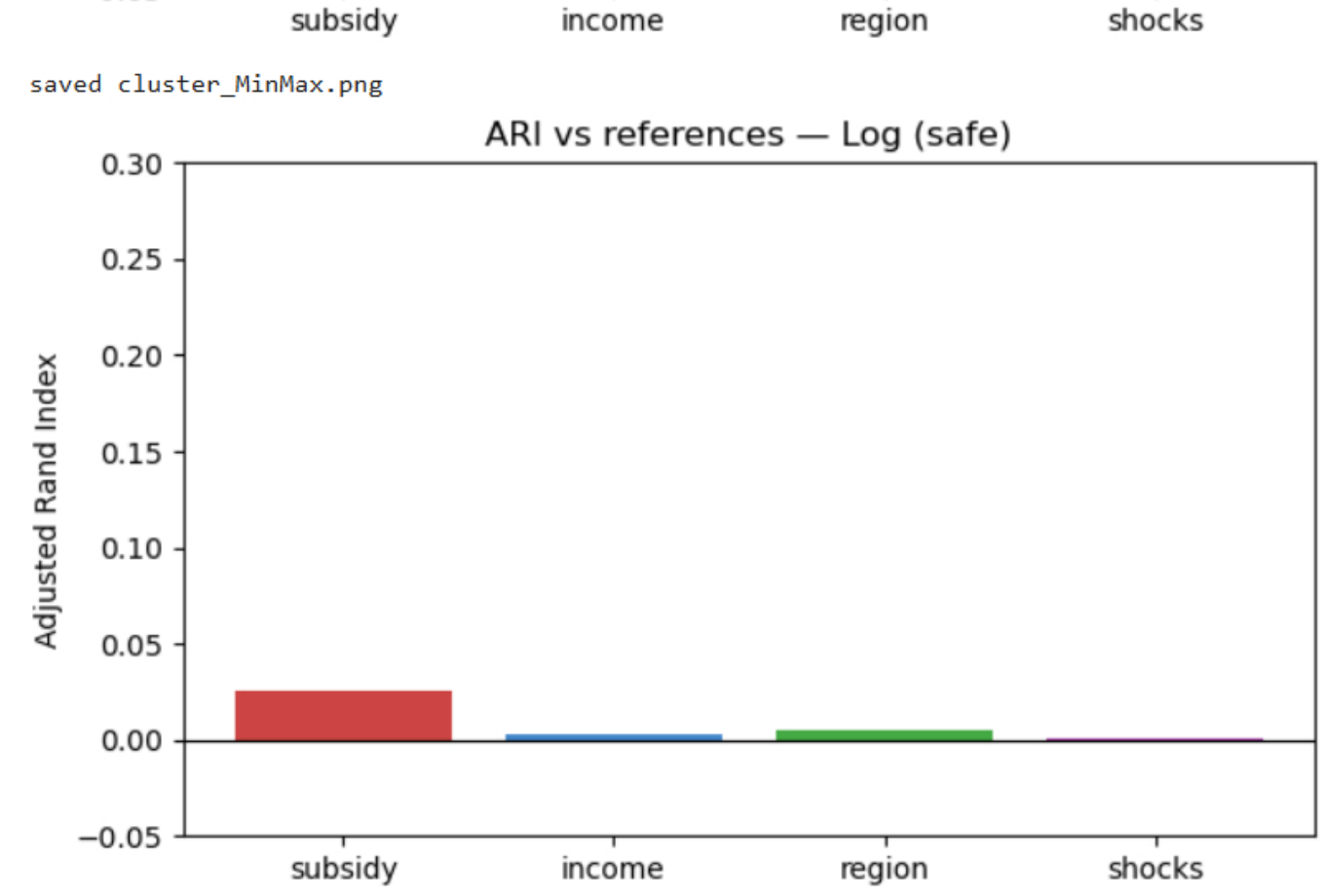
```
saved cluster_Raw.png
```



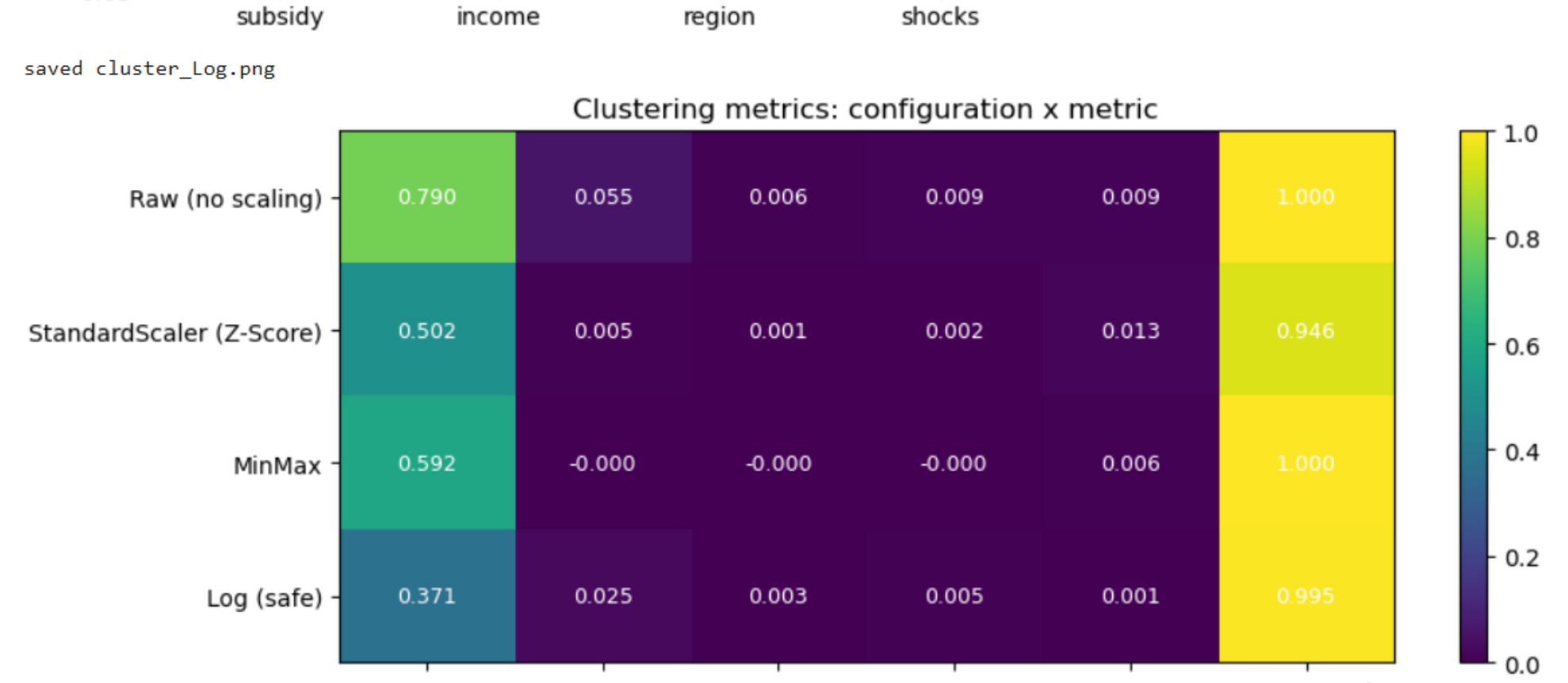
```
saved cluster_StandardScaler.png
```



```
saved cluster_MinMax.png
```



```
saved cluster_Log.png
```



	Silhouette	ARI_subsidy	ARI_income	ARI_region	ARI_shocks	Stability_ARI
Raw (no scaling)	0.790	0.055	0.006	0.009	0.009	1.000
StandardScaler (Z-Score)	0.502	0.005	0.001	0.002	0.013	0.946
MinMax	0.592	-0.000	-0.000	-0.000	0.006	1.000
Log (safe)	0.371	0.025	0.003	0.005	0.001	0.995

```
saved cluster_comparison.png

=== TAKEAWAY (interpretation) ===
1. ARI vs subsidy_level (StandardScaler) = 0.005: modest -> regimes only partly track the subsidy confound.
2. ARI vs shocks (StandardScaler) = 0.013: near zero -> discovered regimes do NOT align with real price shocks.
3. Stability ARI (StandardScaler, 10 seeds) = 0.946 (reproducible).
4. Preprocessing changes the clustering: ARI(raw vs scaled) for StandardScaler = 0.023 (low -> scaling materially changes which weeks cluster together).
```

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